# Multimodal Remote Sensing Image Matching via Learning Features and Attention Mechanism

Yongxian Zhang, Chaozhen Lan, Haiming Zhang, Guorui Ma, and Heng Li

***Abstract*— Matching multimodal remote sensing images (RSIs) remains an ongoing challenge due to the significant nonlinear radiometric differences and geometric distortions, resulting in matches exhibiting one-to-many matches or mismatches. To tackle this challenge, we propose a novel approach for multimodal RSI matching called modality-independent consistency matching (MICM), which leverages the capabilities of deep convolutional neural networks and the transformer attention mechanism to improve the matching performance. The proposed MICM method consists of three key steps. First, a Unet-like feature extraction backbone network is employed to learn multiscale invariant features from multimodal RSIs, enabling the extraction of rich and evenly distributed feature keypoints. Second, a hybrid approach combining local learning features with the transformer attention mechanism is introduced to aggregate learning features, facilitating both detailed capture and long-range modeling to enhance the representation ability of the features. Third, a feature consistency correlation strategy is adopted to maximize the number of correct corresponding feature points, ensuring reliable matching performance. The performance of the proposed method has been extensively evaluated on both the same scene and different scene multimodal RSIs, which are captured from various imaging modes, wavebands, and platforms. The results show the superior matching performance of the proposed MICM method compared to commonly used and state-of-the-art handcrafted- and learning-based methods when evaluated on both the same scene and different scene datasets. The proposed method serves as a valuable reference for addressing common challenges in multimodal RSI matching.**

***Index Terms*— Attention mechanism, feature matching, learning feature, multimodal image, transformer.**

## I. Introduction

THE rapid advancement of Earth observation technology leads to the increased popularity of multimodal remote sensing image (RSI) joint processing. This way does well in integrating RSI time-space-spectrum information to enhance the analysis and interpretation of observation data. In context, multimodal RSI refers to images captured by sensors with diverse imaging mechanisms and characteristics, leading to substantial disparities in the representation of ground objects. Moreover, multimodal RSI encounters large challenges associated with nonlinear radiation and geometry distortions. Precise matching of multimodal RSIs is a crucial task in joint processing, as the matching performance impacts downstream applications. The primary goal of multimodal RSI matching is to establish correct correspondences between two or more images acquired from different platforms, scales, times, and wavebands. By combining their complementary features, the matching enables a more accurate interpretation of the target scene. Now, multimodal RSI matching has found widespread applications in various fields, including image fusion [1], change detection [2], and disaster assessment [3].

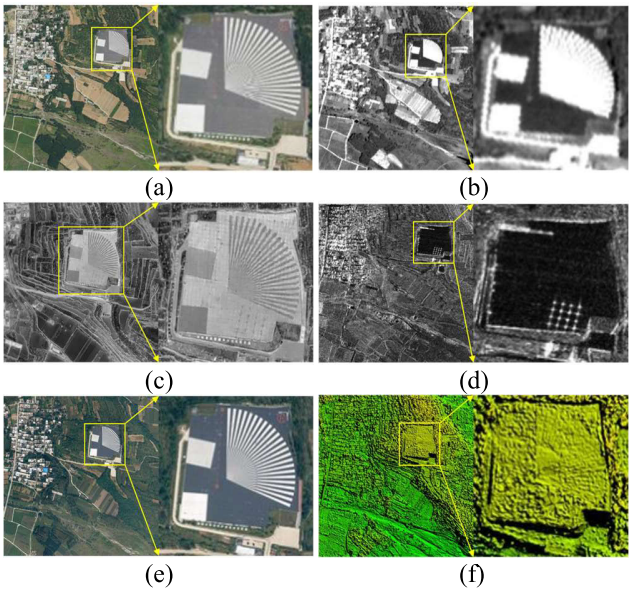


Fig. 1. Multimodal RSIs in the same area. The rectangles are partially enlarged. (a) UAV multispectral image. (b) Ziyuan-3 satellite image. (c) UAV thermal IR image. (d) Gaofen-3 SAR satellite image. (e) DOM. (f) DSM.

As shown in Fig. 1, significant geometric and radiation differences can be perceived in the same regions of images. To realize robust multimodal RSI matching, the key problem is to correct match feature keypoints and eliminate mismatches. For this purpose, a number of image-matching methods have been proposed; they can be roughly summarized into two categories.

1) Handcrafted-based multimodal RSI matching methods mainly rely on the researcher's expertise to design feature detectors, descriptors, and matching strategies

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to achieve efficient matching [4]. According to whether features need to be detected on the reference and sensed images, it can be further divided into two categories: area-based method and feature-based method. The area-based methods first estimate the geometric transformation model by optimizing the similarity measures of predefined templates created from the input two images [5]. These methods have been well applied in the registration of multimodal RSIs with intensity differences [6]; however, it is sensitive to template size, image texture, scale, geometric deformation, and so on [7]. The feature-based methods involve two main steps. First, features are detected, which include point, line, and region features. They serve as distinctive landmarks. Second, a feature-matching strategy is applied to establish correspondences between the detected keypoints in different images.

2) Learning-based methods rely on deep learning technologies to obtain high-level features. The main workflow can be described as an image feature detector to perform local learning from the bottom to the top level, learning robust features and descriptors through multilevel nonlinear mapping to obtain better matches. In the past few years, learning-based methods have been proven powerful in extracting features crucial for multimodal RSI registration [8], [9], [10]. However, challenging applications or recent advances suggest that convolutional neural networks (CNNs) are limited in their ability to understand the spatial correspondence between features. The issue is further exaggerated when it comes to multimodal RSI, in which the appearance of images can differ significantly. These CNN-based methods can capture local context and generate semantic features [11]; however, their global modeling ability is weak to perform long-distance feature information interaction. To take advantage of the stronger representational power of global features and long-distance modeling capability for RSI matching, the researchers introduced the transformer attention mechanism [12]. The core of a transformer model is the self-attention mechanism, which learns weight maps between elements of input sequences. By combining the advantages of CNN and transformer within a unified framework, it is expected to achieve enhanced performance in multimodal RSI matching. This is because this approach can capture both local and global features and yield more precise and robust keypoints, which are important to match multimodal RSIs [13].

As the depth and breadth of RSI matching applications continue to expand, although there have been numerous studies on multimodal matching that have shown promising performance [14], there are still several areas that require further improvement. First, multimodal RSI matching methods can be less adaptable when dealing with large geometric and radiative differences. Therefore, it becomes essential to develop more reliable algorithms that can handle these challenges effectively. The second is the scarcity of publicly accessible multimodal RSI training datasets that limit the effectiveness of feature representation and model performance. This is because access to diverse and comprehensive training data is crucial for training robust and generalizable models. Third, different imaging modes could impact the performance of multimodal RSI matching, making it difficult to obtain corresponding points. This issue arises due to the inherent differences in imaging characteristics and RSI data acquisition modes. For example, optical remote sensing is a passive imaging principle, which is susceptible to cloud cover. Synthetic aperture radar (SAR) remote sensing is an active imaging principle, which can be used for all-day and all-night imaging without the influence of cloud and fog occlusion, but the image contains a lot of random noise and speckle noise, which will blur the structure and details of the image [15], [16]. Finally, some multimodal RSI matching methods mainly match two representative images of different imaging modes (e.g., optical, infrared (IR), and SAR) or from different imaging platforms (e.g., drone, aircraft, and satellite) in multiple classical scenes. The matching performance of multimodal RSIs in the same area (e.g., optical-SAR, optical-IR, and optical-SAR images) lacks generalization, resulting in that the matching performance of multimodal images in the same scene has not been thoroughly verified. It is crucial to utilize multimodal RSIs from the same scene to assess and improve the performance of multimodal RSI matching methods.

Therefore, studying an efficient and reliable multimodal RSI matching method that is applicable to both the same scene and different scene multimodal RSI datasets is essential to better serve joint processing applications. A hybrid approach combining local multiscale learning features and transformer attention mechanism method is proposed to address the issue of one-to-many matches or mismatches in multimodal RSI matching. We conducted extensive evaluations of our method using multimodal RSI captured under both the same and different scenes to assess the matching efficiency and effectiveness. The key contributions are given as follows.

1) We propose a novel multimodal RSI matching method that combines CNN-based local features with the transformer attention mechanism for coupling detail capture and long-range modeling. The proposed method achieves a balance between matching efficiency and accuracy in challenging scenes.
2) We introduce an up- and down-feature detector backbone network to extract multiscale invariant features and a feature keypoint matching transformer to improve feature representation. These strategies effectively enhance the performance of multimodal RSI matching.
3) We only use optical images and a few optical-SAR images as training samples, achieve high-precision matching between optical images and multimodal RSIs, including optical, IR, and SAR under multiplatform and multitime conditions, and verify its superiority over some state-of-the-art multimodal RSI matching methods in terms of accuracy and reliability.

The rest of this article is organized as follows. Section II reviews the related work on CNN- and transformer-based

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image-matching technologies. Section III introduces the proposed multimodal RSI matching method. Section IV shows the experimental dataset, qualitative and quantitative results, and analyses. Section V discusses the factors that influence the efficiency and effectiveness of the proposed method. Section VI provides the study conclusions.

## II. Related Works

Learning-based image matching has become a hot topic for its capabilities in resolving the limitations of traditional image matching [17]. The core utilizes the CNN model as an image high-level feature detector to perform feature learning from the bottom to top levels through multilevel nonlinear mapping. At present, learning-based matching methods can be roughly divided into four categories: image patch-based, detection-then-description, detection-and-description, and transformer-based methods. This section briefly reviews the previous work that is relevant to the proposed method.

### A. Patch-Based Image Matching

The patch-based methods extract features from small image patches using a Siamese network [18] or a triplet network architecture [19]. The goal is to make inliers to be closer and outliers to be far away from each other in the description space [20]. The patch-based model is widely applied in the field of multimodal RSI matching. Merkle et al. [21] proposed to use a Siamese network to learn corresponding features from optical and SAR images. Chen et al. [22] proposed a depth hashes method for multimodal RSIs' matching based on a triplet network model. However, due to the nonlinear geometric, gray, texture, and perspective changes of multimodal RSIs, there are still many difficulties in high-precision matching. To solve these challenges, Tharani et al. [23] developed a residual-dyad descriptor network and aggregated local learning features by means of encoder–decoder architecture, achieving excellent matching performance. In summary, patched-based image-matching models primarily concentrate on how to describe feature blocks efficiently and reliably, and its feature receptive field is limited by block size and lacks global information.

### B. Detection-Then-Description-Based Image Matching

This model first employs a detector to extract informative features and then generates feature descriptors by describing image blocks [24]. This approach is suitable for multimodal RSI matching tasks involving significant radiation distortion. Ye et al. [25] realized multimodal RSI subpixel registration accuracy by the fusion of scale invariant feature transform (SIFT) features and learning features. Zhang et al. [26] adopted a detectable, repeatable, and distinguishable feature detector to achieve high-precision matching. However, when it comes to multimodal RSIs' matching with large geometric and radiation differences, this model may suffer from unstable feature detection. This is attributed to the fact that the feature descriptor is obtained from a large image block, while the feature keypoints are extracted from a small image block, leading to a heavy reliance on low-level information.

### C. Detection-and-Description-Based Image Matching

This model belongs to an end-to-end matching approach that integrates feature detector and description modules, avoiding the drawbacks of multimodule combinations and reducing computational complexity. In the existing studies, Cui et al. [27] embedded features containing high-level semantic information into multimodal RSI matching tasks to achieve IR and optical image matching. However, previous studies [28], [29] have shown that simply increasing the depth of the network does not necessarily lead to improved performance. When utilizing a CNN network to extract features, both high-level semantic features and low-level fine-grained features play an important role in achieving accurate image matching. Zhang et al. [30] explored the multilevel feature fusion technology for multimodal RSI matching. Song et al. [31] improved multimodal RSI matching ability by integrating attention mechanisms. Based on existing literature studies, it is evident that the detection-and-description model plays a crucial role in improving the matching performance of multimodal RSIs.

### D. Transformer-Based Image Matching

Transformer-based matching methods are built upon the attention mechanism via a triplet of key(K), query(Q), and value(V), where K and Q are multiplied by the dot product to obtain the corresponding feature attention weights, and the obtained weights and values are multiplied by the dot product to obtain final information output [32]. SuperGlue [33] utilizes self- and cross-attention in the matching task, leveraging both spatial relationships and the visual appearance of the keypoints, achieving state-of-the-art matching performance. To optimize message passing, SGM-Net [34] proposes a highly sparse graph neural network that makes message passing more efficient. LoFTR [35] utilizes the self-attention layer (SAL) and the cross-attention layer (CAL) to achieve subpixel-level image-matching accuracy by means of a coarse to fine manner. These methods use SAL and CAL to exchange messages across keypoints [36]. In terms of a transformer-based model serving multimodal RSI matching, Li et al. [12] propose to use a wider range of sensitive field and context information to build a cross-model RSI matching network based on self-attention mechanism, and Li et al. [37] propose a pixel-wise mapping model for cross-modality image by integrating self-attention mechanism, achieving optical and SAR image matching. It is evident that the transformer-based structure can extract features including local gray-level information and global context information, and has strong robustness for multimodal image matching.

### E. Relation to Previous Works

The proposed method is closely related to multimodal RSI matching in various real-world scenarios. The difference from the existing multimodal RSI matching method is that we try to incorporate a multiscale invariant feature extraction improved CNN model, transformer-based attention mechanism, and consistency matching strategy to earn rich corresponding points to achieve robust matching performance for multimodal RSI in the same scene and different scene datasets. In contrast to

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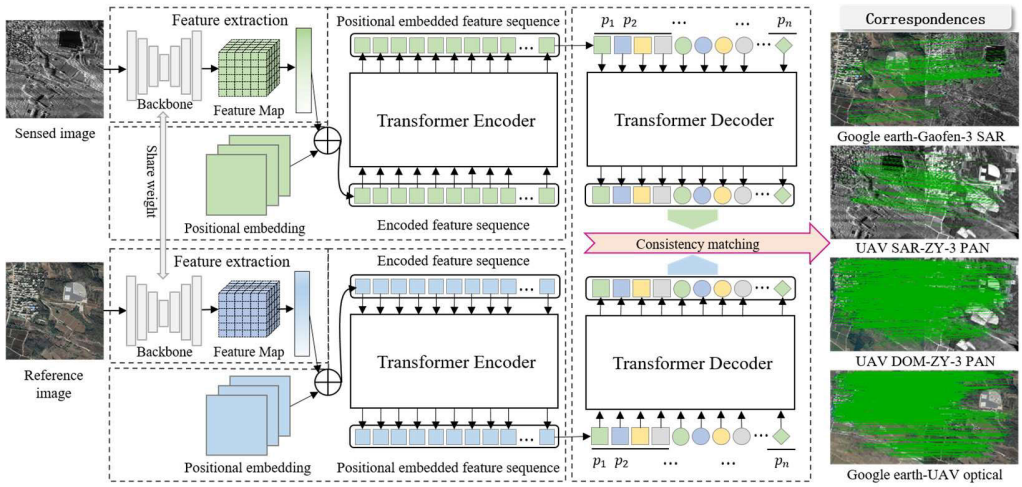


Fig. 2. Workflow of the proposed MICM method.

prior methods, our method combines local multiscale learning features with the attention mechanism, aiming to make feature descriptors "look wider to describe better" [38]. In addition, previous methods have mainly concentrated on matching specific types of multimodal RSIs, whereas our method addresses the requirements of matching multimodal RSIs in complex scenes, both within the same and different scenes.

## III. Methodology

In this section, we will introduce a novel multimodal RSI matching method that consists of a feature extraction module, a transformer encoder module ($\mathcal{T}_{\mathcal{E}}$), a transformer decoder module ($\mathcal{T}_{\mathcal{D}}$), and a consistency matching module [39], as shown in Fig. 2. The feature extraction module is used to obtain multiscale local feature maps of the input image pairs $I_A$ and $I_B$. In this step, each branch is a hybrid structure consisting of a Unet-like backbone extracting CNN local features and a $\mathcal{T}_{\mathcal{E}}$ modeling global context from the CNN features and a $\mathcal{T}_{\mathcal{D}}$ yield initial corresponding keypoints. Finally, a consistency correlation strategy is used to obtain high-precision corresponding keypoints. From this pipeline, we can obtain rich matches on the same and different scene datasets.

### A. Robust Learning Feature Extraction

Given the reference and sensed images $I_A$ and $I_B$, our goal is to extract the abundant learning features $F_A$ and $F_B$ and match them to yield a set of correspondences $M_{A\leftrightarrow B}$. To achieve this goal, we use a CNN model that is similar to a U-net, as a backbone to extract multiscale feature maps to remedy the potential scale variance from the input multimodal RSI pairs in the feature detector module, as shown in the left-hand side of Fig. 2. This detector backbone network model enables efficiently computing a keypoint map by downsample/upsample operations in the original image resolution. Among them, spatial downsampling is performed first to extract the high-level semantic information, and then, upsampling is performed to recover the spatial information, and low-level information and high-level information are fused by skip connections [40]. In our feature extraction experiment, we employ four downblocks for the encoder and 4 upblocks for the decoder, with skip connections, each block having two convolutional layers with $5 \times 5$ kernels followed by rectified linear unit (ReLU) and batch normalization. After obtaining feature maps, we represent them as $F(I_A)$ and $F(I_B)$, respectively, and concatenate feature maps along the spatial dimension to form $F = \{F(I_A), F(I_B)\}$.

To obtain high responses in confident regions for feature detection and matching, we divide feature maps into grids of size $g_k \times g_k$ and select at most one point from each grid cell based on feature map scores in this cell as the feature keypoints. As introduced in discrete keypoints (DISK) [41], we utilize a probability distribution $P_{kp}$ over each grid cell based on the heat map scores in each grid cell, which is used to determine candidate feature keypoints $P_A = \{x_{A1}, x_{A2}, \ldots\}$ and $P_B = \{x_{B1}, x_{B2}, \ldots\}$, respectively. Here, the probabilistic framework is denoted as $P_s(p|K^u) = \text{softmax}(K^u)_p$, where $p$ is the pixel, $u$ is the cell, and $K$ is the feature map. Once feature locations are known, we associate them with the $l_2$ normalized descriptors at this location, yielding a set of output features. In this stage, we adopt the nonmaximum suppression (NMS) to obtain reliable features. Next, feature maps are converted into 1-D sequential data in the form of $d \times HW$ because $\mathcal{T}_{\mathcal{E}}$ takes sequential data as input. Therefore, the feature position embedding can be used to add the feature information before sending it to each attention layer as input. Finally, we concatenate the feature map for the two corresponding images side by side and add the positional encoding $\mathcal{P}$ of the feature keypoints coordinate function $\Omega$ to produce a new feature map $\mathbf{F}$, written as

$$\mathbf{F} = [F(I_A), F(I_B)] + P(\Omega). \tag{1}$$

To explore a suitable NMS window size for multimodal RSIs, we test NMS window size from $1 \times 1$, $3 \times 3$, $5 \times 5$, $7 \times 7$, $9 \times 9$, and $11 \times 11$ to ensure that the keypoints are abundant and evenly distributed in the reference and sensed image; Fig. 3 shows feature keypoints detection results of the ZY3-PAN, unmanned aerial vehicle (UAV)-IR, and GF3-SAR

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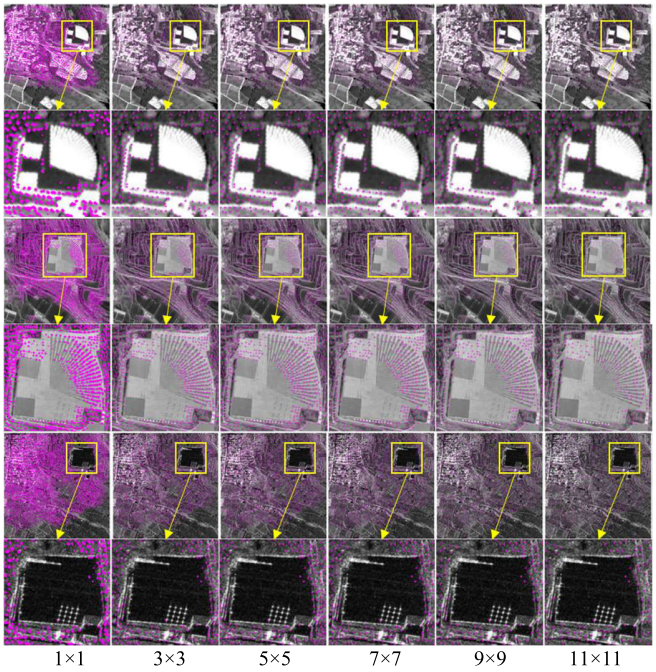


Fig. 3. Learning feature detection results in different imagining mechanisms [from (top) to (bottom): optical (ZY3-PAN), IR (UAV-IR), and SAR (GF3-SAR)] with the different NMS window sizes (from (left) to (right): $1 \times 1$, $3 \times 3$, $5 \times 5$, $7 \times 7$, $9 \times 9$, and $11 \times 11$).

sensed RSIs based on the proposed learning feature detector. Several conclusions are reached from the detection results.

1) The feature detector is insensitive to images with different imaging modes and scales, and can detect rich feature keypoints in optical, IR, and SAR RSIs.
2) The window size of the NMS has an obvious impact on the number of detected feature points. The smaller the window size, the more the feature points, and the larger the window size, the fewer the feature points.
3) There are obvious differences in the number of features detected by smooth large buildings on different imaging mode images, and the number of feature detection results is not affected by the NMS window size, such as the white area on ZY3-PAN image, the gray area on UAV-IR image, and the black area on GF3-SAR image.
4) The detection features have rich quantity and good distribution uniformity, which is conducive to feature matching.

Based on the findings from the above results, it is evident that the discriminability of the local CNN descriptor is generally limited, which may lead to mismatches. To address this issue, it is necessary to incorporate a feature enhancement module to obtain more discriminative representations for each feature keypoint. Inspired by self-/cross-attention mechanism-based feature enhancement methods, we introduce a novel approach called the feature keypoints matching transformer (FKMT) in Sections III-B–III-D. The FKMT is specifically designed for enhancing feature learning, aiming to improve the performance of feature matching. It includes three modules: linear positional embedding, transformer encoder, and transformer decoder.

### B. Linear Positional Embedding

To make full use of the feature location information, we add the positional embeddings with the features output from SAL and CAL. We utilize a learnable linear positional embedding, similar to the approach introduced in [42], to embed features obtained from the Unet-like backbone network. This involves adding the positional embedding to the image feature patch, which allows the transformed CNN features to become position-dependent. For a given normalized feature location $\boldsymbol{x} = [x, y] \in [0, 1]$, we write as

$$\mathcal{P}(\boldsymbol{x}) = \left[p_1(\boldsymbol{x}), p_2(\boldsymbol{x}), \ldots, p_{\frac{N}{4}}(\boldsymbol{x})\right] \tag{2}$$

$$p_k(\boldsymbol{x}) = \left[\sin\left(k\pi x^{\mathrm{T}}\right), \cos\left(k\pi x^{\mathrm{T}}\right)\right] \tag{3}$$

where $N = 128$ is the number of channels. $p_k$ generates four values as the output of encoder $P$ is size $N$.

### C. Transformer Encoder

The transformer-based method enables the model to capture and utilize information from different correspondences by leveraging a self-attention mechanism that measures the relevance of input sequences using the Query($\mathbf{Q}$)-Key($\mathbf{K}$)-Value($\mathbf{V}$) model [13]. In multimodal RSI matching, the significant domain gap between different modalities poses challenges in embedding image features into a shared space using a domain-shared network [43]. To address this problem, we propose a Siamese-like network that learns separate representations for the reference and sensed images. These representations are then projected onto a $\mathcal{T}_{\mathcal{E}}$ to share feature space, facilitating multimodal RSI matching. Here, $\mathcal{T}_{\mathcal{E}}$ encodes tokenized image patches from a CNN feature map as the input sequence for extracting global contexts. In contrast to nonlocal blocks that compute spatial attention within a single image, our approach utilizes CAL to establish spatial correspondences between two images of different modalities [44]. Our framework employs two six-layer cross-modal transformers to yield a coattention representation of multimodal RSIs. Self-attention and cross-attention mechanisms are utilized to aggregate long-range context information within and between images, respectively. The SAL aims to learn reliable relationships between keypoints within each feature set, while the CAL computes cross-dependencies between the two feature sets to reconstruct the feature representations and establish correspondences.

In the transformer encoder, $\mathcal{T}_{\mathcal{E}}$ is made up of $N$ sequential encoder blocks, each of which contains two sublayers: a multihead SAL and a feedforward network layer. The $\mathcal{T}_{\mathcal{E}}$ components are presented in the left-hand side of Fig. 4. Both sublayers employ LayerNorm, followed by a residual connection, to facilitate effective information flow. Since the transformer is permutation-invariant, we supplement it with fixed positional encodings that are added to the input of each attention layer [45]. The multihead SAL focuses on capturing relationships within a single image by retrieving relevant information from different parts. This intra-image attention mechanism allows for long-range global aggregation, enabling the model to capture dependencies and correlations between

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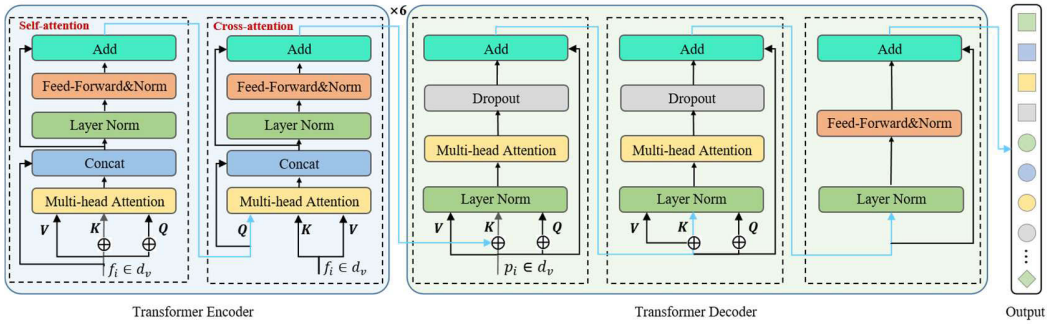


Fig. 4. Transformer encoder ($\mathcal{T}_{\mathcal{E}}$ and transformer decoder ($\mathcal{T}_{\mathcal{D}}$ architecture.

different regions. On the other hand, when the **Q** and **K** vectors are from different views, the multihead attention layers facilitate cross-relationships across these two views. Therefore, this interimage attention mechanism enables feature interaction and correlation. By attending to relevant information, the model can establish connections and capture cross-modal dependencies, enhancing matching performance. Moreover, the feedforward network layer in each encoder block consists of fully connected layers, which helps in further refining the feature representations and extracting higher level information.

The multihead SAL is a critical component of $\mathcal{T}_{\mathcal{E}}$. The input feature map $\mathbf{F} \in R^{N\times D}$ is encoded by $W$ to yield three matrix representations: $\hat{\mathbf{Q}}_S = \mathbf{F}\mathbf{W}_i^Q$, $\hat{\mathbf{K}}_S = \mathbf{F}\mathbf{W}_i^K$, and $\hat{\mathbf{V}}_S = \mathbf{F}\mathbf{W}_i^V$, where $\mathbf{W}_i^Q \in R^{d_f\times d_k}$, $\mathbf{W}_i^K \in R^{d_f\times d_k}$, and $\mathbf{W}_i^V \in R^{d_f\times d_v}$ denote learnable weights for feature transformation, and the subscript $i$ denotes the $i$th attention head in SAL. Then, the attention-weighted value is computed via (4). Here, $\hat{\mathbf{Q}}_S$ is the salient feature map, $\hat{\mathbf{K}}_S$ is the matrix of feature keypoints grid coordinates, and $\hat{\mathbf{V}}_S$ is the score map. The weight is produced by "scaled dot-product attention" strategy

$$\mathbf{A} = \text{softmax}\left(\frac{\hat{\mathbf{Q}}_S\hat{\mathbf{K}}_S^T}{\sqrt{d_k}}\right) \tag{4}$$

$$\text{Attention}\left(\hat{\mathbf{Q}}_S, \hat{\mathbf{K}}_S, \hat{\mathbf{V}}_S\right) = \mathbf{A}\hat{\mathbf{V}}_S \tag{5}$$

where **A** is the attention weight matrix, and each element of **A** represents the pairwise similarity between two elements of the input sequence and their query and key representations. $\mathbf{A}\hat{\mathbf{V}}_S$ is used to compute a normalized score for each input token based on the dot product of the Query and Key representations. This score is used to weight the corresponding Value representation of the token, indicating to the network whether to focus on or ignore it. We compute the values for all heads and concatenate them together with equations

$$\text{head}_i = \text{Attention}\left(\mathbf{F}\mathbf{W}_i^Q, \mathbf{F}\mathbf{W}_i^K, \mathbf{F}\mathbf{W}_i^V\right) \tag{6}$$

$$\text{MultiHead}(\mathbf{F}) = \text{concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)\mathbf{W}^S \tag{7}$$

where $\mathbf{W}^S \in R^{hd_v\times d_f}$ and $h$ is the number of heads in SAL. In our experiment, the number of attention heads in multihead self-attention is set to 6.

The feedforward network layer uses a linear transformation followed by a nonlinear activation function for each of the weighted sums. To further adjust feature representations, the position of the feedforward layer transforms each feature map with three fully connected layers. We write as

$$\text{FFN}(x) = \text{ReLU}(\text{ReLU}(x\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2)\mathbf{W}_3 + b_3 \tag{8}$$

where $x \in R^{1\times d_f}$, $\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3 \in R^{d_f\times d_f}$, and $b_1, b_2, b_3 \in R^{1\times d_f}$.

Different from the SAL, CAL aims to enhance the keypoints attributes in image $I_A$ using the information from the counterpart image $I_B$ [46]. For the image-matching problem, it is necessary to conduct cross-learning on both the reference and sensed images to achieve information communication on two input images [47]. This is because combining two attention mechanisms can promote the representation power of the extracted features [48], [49]. To achieve interactive information propagation from the sensed to reference points, we need to initial the queries, keys, and values as $\tilde{\mathbf{Q}}_C$, $\tilde{\mathbf{K}}_C$, and $\tilde{\mathbf{V}}_C$ by feature map **F**. For the reference image, query, key and value matrices are represented as $\mathbf{Q}^A$, $\mathbf{K}^B$, and $\mathbf{V}^B$, respectively. For the sensed image, query, key, and value matrices are denoted as $\mathbf{Q}^B$, $\mathbf{K}^A$, and $\mathbf{V}^A$. The cross-attention module takes the different $\mathbf{K}_C$ and $\mathbf{V}_C$ of the reference image and sensed image as the input. Similar to SAL, CAL employs a multihead architecture to project the input attributes into different latent spaces. Here, the query, key, and value are formed with the following equations:

$$\tilde{\mathbf{Q}}_C = \mathbf{F}\mathbf{W}_j^Q = \mathbf{Q}_C\tilde{\mathbf{W}}_j^Q \tag{9}$$

$$\tilde{\mathbf{K}}_C = \mathbf{F}\mathbf{W}_j^K = \mathbf{K}_C\tilde{\mathbf{W}}_j^K \tag{10}$$

$$\tilde{\mathbf{V}}_C = \mathbf{F}\mathbf{W}_j^V = \mathbf{V}_C\tilde{\mathbf{W}}_j^V. \tag{11}$$

Subsequently, the cross-attentions between the queries and keys are computed as

$$\text{Attention}\left(\tilde{\mathbf{Q}}_C, \tilde{\mathbf{K}}_C, \tilde{\mathbf{V}}_C\right) = \text{softmax}\left(\frac{\tilde{\mathbf{Q}}_C\tilde{\mathbf{K}}_C^T}{\sqrt{d_k}}\right)\tilde{\mathbf{V}}_C. \tag{12}$$

Similar to SAL, we transform the concatenation of the outputs of attention heads through a learnable parameter matrix $\mathbf{W}^C \in R^{hd_v\times d_f}$ as

$$\text{head}_i = \text{Attention}\left(\mathbf{Q}_C\tilde{\mathbf{W}}_j^{\mathbf{Q}}, \mathbf{K}_C\tilde{\mathbf{W}}_j^{\mathbf{K}}, \mathbf{V}_C\tilde{\mathbf{W}}_j^{\mathbf{V}}\right) \tag{13}$$

$$\text{MultiHead}(\mathbf{F}) = \text{concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)\mathbf{W}^C \tag{14}$$

where $h$ is the number of attention heads in the CAL. Finally, we update the keypoint attributes using a residual learning scheme

$$\text{FFN}(y) = \text{ReLU}(\text{ReLU}(y\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2)\mathbf{W}_3 + b_3 \tag{15}$$

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where $y \in R^{1\times d_f}$, $\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3 \in R^{d_f\times d_f}$, and $b_1, b_2, b_3 \in R^{1\times d_f}$.

### D. Transformer Decoder

$\mathcal{T}_\mathcal{D}$ follows the standard architecture of the transformer, transforming $N$ embeddings of size $d$ using multihead attention mechanisms. The difference with the original transformer is that our model decodes the $N$ objects in parallel at each decoder layer. The detail components of $\mathcal{T}_\mathcal{D}$ are illustrated in the right-hand side of Fig. 4. The goal of $\mathcal{T}_\mathcal{D}$ is to decode the encoded feature keypoints position information from the input image to the coarse corresponding position on the reference image. To maintain the independence between queries, self-attention is disabled in the decoder. This ensures that each query can focus exclusively on the relevant information extracted from the encoded images. By disallowing self-attention within the decoder, queries can undergo effective and individualized decoding, free from influence from other queries, similar to the decoder structures used in DETR [45] and VisTR [50]. It can offer two benefits: 1) generating more robust coarse- and fine-grained features to improve the final matching performance and 2) designing a lightweight decoder that maintains computational efficiency without increasing the overall complexity of the model. In the experiment, $\mathcal{T}_\mathcal{D}$ follows the standard transformer architecture and utilizes a multihead attention mechanism to embed $N$ features of size $d$ into $\mathcal{T}_\mathcal{D}$. After processing by the decoder, the corresponding feature points are described by a 128-d vector. Subsequently, a three-layer multilayer perceptron (MLP) to regress the pixel coordinates of the corresponding keypoints, yielding the coarse position information on the reference image. The designed decoding structure consists of six layers, and each layer of the decoder module contains six heads.

Algorithm 1 displays the FKMT process via the transformer self- and cross-attention mechanisms.

**Algorithm 1** Transformer Attention Mechanism Module

**Input:** Feature keypoints in image pair $(I_A, I_B)$;
Positional embeddings $\mathcal{P}(\mathbf{\Omega})$;
The number of attention layers $N_L$;
The numbers of self- and cross-attention heads $N_S$, $N_C$;
Learnable parameters in SAL $\hat{\mathbf{W}}_i^Q, \hat{\mathbf{W}}_i^K, \hat{\mathbf{W}}_i^V, \boldsymbol{W}^S$;
Learnable parameters in CAL $\tilde{\mathbf{W}}_j^Q, \tilde{\mathbf{W}}_j^K, \tilde{\mathbf{W}}_j^V, \boldsymbol{W}^C$
**Output:** Corresponding keypoints $M_{A\leftrightarrow B}$
**Procedure** Feature keypoints matching transformer
1: **for** $t= 1; t < N_L; t++$ do
*//self-attention layer*
2: $\mathbf{F}^A \rightarrow \mathbf{Q}^A, \mathbf{K}^A, \mathbf{V}^A$; $\mathbf{F}^B \rightarrow \mathbf{Q}^B, \mathbf{K}^B, \mathbf{V}^B$
3: **for** $i = 1$; $i \le N_S$; $i++$ do
4: $\hat{\mathbf{Q}}_i^{\mathbf{A}} = \mathbf{Q}^A\hat{\mathbf{W}}_i^Q, \hat{\mathbf{K}}_i^{\mathbf{A}} = \mathbf{K}^A\hat{\mathbf{W}}_i^K, \hat{\mathbf{V}}_i^{\mathbf{A}} = \mathbf{V}^A\hat{\mathbf{W}}_i^V$;
$\hat{\mathbf{Q}}_i^{\mathbf{B}} = \mathbf{Q}^B\hat{\mathbf{W}}_i^Q, \hat{\mathbf{K}}_i^{\mathbf{B}} = \mathbf{K}^B\hat{\mathbf{W}}_i^K, \hat{\mathbf{V}}_i^{\mathbf{B}} = \mathbf{V}^B\hat{\mathbf{W}}_i^V$;
5: $\hat{\boldsymbol{F}}_i^{\boldsymbol{A}} = \text{softmax}\left(\frac{\hat{\boldsymbol{Q}}_i^A(\hat{\boldsymbol{K}}_i^A)^T}{\sqrt{d_k}}\right)\hat{\boldsymbol{V}}_i^{\boldsymbol{A}}$; $\hat{\boldsymbol{F}}_i^{\boldsymbol{B}} = \text{softmax}\left(\frac{\hat{\boldsymbol{Q}}_i^B(\hat{\boldsymbol{K}}_i^B)^T}{\sqrt{d_k}}\right)\hat{\boldsymbol{V}}_i^{\boldsymbol{B}}$
6: **end for**
7 $\hat{\mathbf{F}}^{\mathbf{A}} = \text{concat}(head_1, head_2, \ldots, head_h)\boldsymbol{W}^S$;
$\hat{\mathbf{F}}^{\mathbf{B}} = \text{concat}(head_1, head_2, \ldots, head_h)\boldsymbol{W}^S$
8: $\mathbf{F}^A = \text{ReLU}(\text{ReLU}(x\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + b_2)\boldsymbol{W}_3 + b_3$;
$\mathbf{F}^B = \text{ReLU}(\text{ReLU}(x\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + b_2)\boldsymbol{W}_3 + b_3$
*//cross-attention layer*
9: $\mathbf{F}^A \rightarrow \mathbf{Q}^A, \mathbf{K}^B, \mathbf{V}^B$; $\mathbf{F}^B \rightarrow \mathbf{Q}^B, \mathbf{K}^A, \mathbf{V}^A$
10: **for** $j = 1$; $j \le N_C$; $j++$ do
11: $\tilde{\boldsymbol{Q}}_j^{\boldsymbol{A}} = \boldsymbol{Q}^A\tilde{\boldsymbol{W}}_j^Q, \tilde{\boldsymbol{K}}_j^{\boldsymbol{A}} = \boldsymbol{K}^A\tilde{\boldsymbol{W}}_j^K, \tilde{\boldsymbol{V}}_j^{\boldsymbol{A}} = \boldsymbol{V}^A\tilde{\boldsymbol{W}}_j^V$;
$\tilde{\boldsymbol{Q}}_j^{\boldsymbol{B}} = \boldsymbol{Q}^B\tilde{\boldsymbol{W}}_j^Q, \tilde{\boldsymbol{K}}_j^{\boldsymbol{B}} = \boldsymbol{K}^B\tilde{\boldsymbol{W}}_j^K, \tilde{\boldsymbol{V}}_j^{\boldsymbol{B}} = \boldsymbol{V}^B\tilde{\boldsymbol{W}}_j^V$;
12: $\tilde{\boldsymbol{F}}_j^{\boldsymbol{A}} = \text{softmax}\left(\frac{\tilde{\boldsymbol{Q}}_j^A\left(\tilde{\boldsymbol{K}}_j^A\right)^T}{\sqrt{d_k}}\right)\tilde{\boldsymbol{V}}_j^{\boldsymbol{A}}$; $\tilde{\boldsymbol{F}}_j^{\boldsymbol{B}} = \text{softmax}\left(\frac{\tilde{\boldsymbol{Q}}_j^B\left(\tilde{\boldsymbol{K}}_j^B\right)^T}{\sqrt{d_k}}\right)\tilde{\boldsymbol{V}}_j^{\boldsymbol{B}}$
13: **end for**
14: $\tilde{\mathbf{F}}^{\boldsymbol{A}} = \text{concat}(head_1, head_2, \ldots, head_h)\boldsymbol{W}^C$;
$\tilde{\mathbf{F}}^{\boldsymbol{B}} = \text{concat}(head_1, head_2, \ldots, head_h)\boldsymbol{W}^C$
15: $\mathbf{F}^A = \text{ReLU}(\text{ReLU}(y\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + b_2)\boldsymbol{W}_3 + b_3$;
$\mathbf{F}^B = \text{ReLU}(\text{ReLU}(y\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + b_2)\boldsymbol{W}_3 + b_3$
16: **end for**
17: $\mathcal{T}_\mathcal{D} \rightarrow$ Coarse corresponding keypoints $M_{A\leftrightarrow B}$
**end procedure**

### E. Loss Function

Our loss function includes two parts: coarse matching loss and symmetric consistency fine-matching loss to supervise the model training. Consider an image pair $(I_a, I_\text{b})$, given the sampled feature keypoint set $\boldsymbol{x} = [x, y]$ on the sensed image $I_a$. The corresponding points are represented as $\boldsymbol{x}' = [x', y']$ on the reference image $I_\text{b}$. Corresponding keypoint relation can be regarded as parameters $\Phi$ optimization of correlation function $\mathcal{F}_\Phi(\boldsymbol{x}|I_A, I_B)$ and denoted as

$$\mathcal{L}_\text{cor} = \left\|\boldsymbol{x}' - \mathcal{F}_\Phi(\boldsymbol{x}|I_A, I_B)\right\|_2^2. \tag{16}$$

Due to the large differences in geometry and radiation, sharing a common query for paired images is not sufficient. To provide consistency supervision, we introduce a symmetric consistency loss, which ensures forward–backward mapping of the correspondences to be spatially close. Consistency loss is commonly employed in feature-matching pipelines [51]. For image pair $(I_a, I_\text{b})$, the output $(f_a, f_\text{b})$ of feature correlation is embedded with $\mathcal{T}_\mathcal{D}$ output $(q_a, q_\text{b})$, as shown in Fig. 2. We further embed $(q_a, q_\text{b})$ to $(f_b, f_\text{a})$, respectively, for correspondence consistency

$$\mathcal{L}_\text{con} = \sum_{i=a}^{b}\left(\lambda_\text{con}\left\|p_i - \hat{p}_i\right\|_2^2 + \lambda_\text{loc}\left\|p_i - \bar{p}_i\right\|_2^2\right) \tag{17}$$

where $p_i$, $\hat{p}_i$, and $\bar{p}_i$ represent the ground-truth, prediction, and symmetric consistency feature points, respectively. $\lambda_\text{con}$ and $\lambda_\text{loc}$ are hyperparameters to balance losses.

The final loss is composed of coarse matching loss $\mathcal{L}_\text{cor}$ and fine-matching loss $\mathcal{L}_\text{con}$, which is formulated as

$$\mathcal{L}_\text{total} = \mathcal{L}_\text{cor} + \mathcal{L}_\text{con}. \tag{18}$$

### F. Implementation

*1) Datasets:* we use a subset of the MegaDepth dataset [52], from which we choose 135 scenes with 63k images total, to train our network model. These images are posed with the COLMAP software. COLMAP provides dense depth estimates, which we utilize to establish pixel-to-pixel correspondences. The depth map is considered as ground truth. Furthermore, we conducted fine-tuning using a subset of the

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SEN1-2 dataset [53] to enhance the robustness of the model performance. The SEN1-2 dataset comprises 282 384 pairs of corresponding SAR and optical imagery patches, collected from across the globe and throughout all meteorological seasons.

*2) Training:* Experiments are performed on an NVIDIA RTX 3090 GPU and 24-GB RAM. For the feature extraction module, we use a four-up- and down-U-Net architecture initialized as our backbone model with pretrained weight. For the transformer architecture, we set six layers to encode local features. Each $\mathcal{T}_{\mathcal{E}}$ layer contains an SAL with six heads. Then, we use a three-layer MLP to convert the transformer output, with a 128-d feature description, followed by ReLU activation. Our model is trained in two stages. First, we freeze the pretrained backbone and only train the remaining parts of the network. This stage consists of 50k iterations, with the SGD optimizer, a learning rate of $10^{-5}$, and a batch size of 24. Then, we unfreeze the backbone and fine-tune the entire model end-to-end. This fine-tuning stage lasts for 20k iterations with a learning rate of $10^{-5}$ and a batch size of 24.

## IV. Experiments

In this section, we begin by providing an overview of the experimental dataset, followed by a description of evaluation metrics and a comparison of the methods used for matching performance assessment. Subsequently, we present detailed experiments and analyses conducted on each dataset.

### A. Experimental Datasets

To assess the generalization capability of the proposed modality-independent consistency matching (MICM), we conduct rich experiments using two challenging multimodal RSI datasets. These test datasets exhibit large nonlinear differences in geometry and radiation, providing a rigorous evaluation of the effectiveness. The first, the same scene dataset, focuses on a specific domain where RSIs are acquired from a single scene over different imaging sensors and modes. The second, different scene dataset, was collected from various scenes. These RSIs exhibit large variations in terms of sensor types, resolutions, viewpoints, and environmental conditions. The dataset covers a wide range of geographic locations and includes scenes with complex topography, urban areas, and natural landscapes. Both the same scene dataset and the different scene dataset consist of 18 pairs of multimodal RSIs, with large geometric and radiative differences. Therefore, these datasets serve as appropriate platforms to evaluate the robustness and reliability of the proposed method.

### B. Evaluation Metrics

To evaluate the performance of the proposed method, we conduct a comprehensive set of qualitative and quantitative experiments. The qualitative evaluation involves analyzing the number of matching points and the distribution of feature points. This assessment allowed us to visually observe the effectiveness of the model-matching process. For quantitative evaluation, we employ mathematical modeling techniques to assess the accuracy and reliability of the matching results. These evaluations provided a comprehensive assessment of the proposed method's performance. We use the number of initial matches (NIM), the number of correct matches (CM), and the root mean square error (RMSE) as quantitative evaluation metrics. A correct match represents a correspondence whose projection error is less than 3 pixels compared with the ground truth. We report the average results of three runs to ensure the reliability of our experiments. CM illustrates the ability of the method to match features, and RMSE shows the error fluctuation of the proposed method in locating feature points, respectively. The formula of RMSE is given as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\left(x_i^1 - x_i^2\right)^2 + \left(y_i^1 - y_i^2\right)^2\right)} \tag{19}$$

where $N$ is the number of matching points; $x_i^1, x_i^2$ and $y_i^1, y_i^2$ are the coordinates of the correspondence on the sensed image and the reference image, respectively.

### C. Comparative With the State-of-the-Art Methods

Six commonly used and state-of-the-art methods are chosen to compare with the proposed method. There are handcrafted-based RIFT [54] and ASS [5] and learning-based D2Net [55], CMMNet [56], ECOTR [57], and LightGlue [58]. To sum up, RIFT uses phase congruency for feature point detection and maximum index map for feature description. ASS uses local nonmaximum on the minimum self-similarity map for feature detection and distribution histogram of the index map for feature description. D2Net is a describe-and-detect approach to conduct sparse local feature detection and description, where feature detection is not done on low-level image structures but postponed until more reliable information is available. CMMNet is an improved feature-matching method for heterogeneous RSIs. It aims to learn pixel-level correlations for geometric consistency in pairs of input images. ECOTR is a coarse-to-fine transformer-based network for local feature matching, which is stagewisely connected to refine the predicted coordinates upon a shared multiscale feature extraction network. LightGlue aims to maintain matching performance while significantly improving inference speed. To make a fair comparison, all experiments were set with recommended parameters.

### D. Experiments on the Same Scene Dataset

*1) Description of the Same Scene Dataset:* To validate the effectiveness of the proposed algorithm, we first conduct qualitative and quantitative comparative experiments on the same scene dataset, as described in Table I. This dataset was collected from images captured on the same surface but with different imaging modes and imaging platforms at different times. Detailed information regarding the image data source, band type, pixel size, and resolution is provided. This dataset was acquired from various sensors, at different times, resolutions, scales, and textures, using different satellite and UAV platforms. The diversity in this dataset allows for comprehensive testing and evaluation of the performance of the proposed algorithm under different conditions and scenarios.

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TABLE I
EXPERIMENTAL DATA DESCRIPTION

| No. | Source | Pixel | Resolution(m) | Characteristic |
|---|---|---|---|---|
| 1 | Google Earth (T1) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 1.0 / 2.0 | The reference image is Google Earth (T1), covering a hill area with an image calibration field and a village, which differs from the sensed image in terms of imaging mode, texture, and scale |
| 2 | Google Earth (T1) / UAV (IR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 3 | Google Earth (T1) / Gaofen-3 (SAR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 4 | Google Earth (T2) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 1.0 / 2.0 | The reference image is Google Earth (T2), which differs significantly from the sensed image in terms of imaging mode, imaging waveband, and scale |
| 5 | Google Earth (T2) / UAV (IR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 6 | Google Earth (T2) / Gaofen-3 (SAR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 7 | UAV (OPT) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 0.5 / 2.0 | The reference image is UAV (OPT), which differs significantly from the sensed image in terms of imaging mode, imaging waveband, and scale |
| 8 | UAV (OPT) / UAV (IR) | 512×512 / 512×512 | 0.5 / 1.0 | |
| 9 | UAV (OPT) / Gaofen-3 (SAR) | 512×512 / 512×512 | 0.5 / 1.0 | |
| 10 | UAV (SAR) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 1.0 / 2.0 | The reference image is UAV (SAR), which differs significantly from the sensed image in terms of imaging mode, imaging waveband, and scale |
| 11 | UAV (SAR) / UAV (IR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 12 | UAV (SAR) / Gaofen-3 (SAR) | 512×512 / 512×512 | 1.0 / 1.0 | |
| 13 | UAV (DOM) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 0.5 / 2.0 | The reference image is UAV (DOM), which differs significantly from the sensed image in terms of imaging mode, imaging waveband, and scale |
| 14 | UAV (DOM) / UAV (IR) | 512×512 / 512×512 | 0.5 / 1.0 | |
| 15 | UAV (DOM) / Gaofen-3 (SAR) | 512×512 / 512×512 | 0.5 / 1.0 | |
| 16 | UAV (DSM) / Ziyuan-3 (PAN) | 512×512 / 512×512 | 0.5 / 2.0 | The reference image is UAV (DSM), which differs significantly from the sensed image in terms of imaging mode, imaging waveband, and scale |
| 17 | UAV (DSM) / UAV (IR) | 512×512 / 512×512 | 0.5 / 1.0 | |
| 18 | UAV (DSM) / Gaofen-3 (SAR) | 512×512 / 512×512 | 0.5 / 1.0 | |

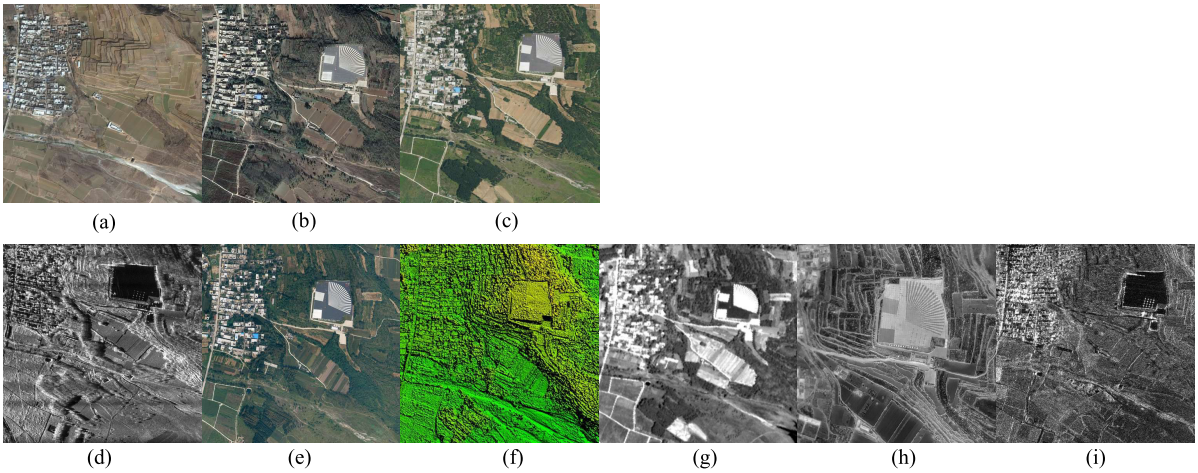


Fig. 5. Multimodal RSIs of the same scene dataset (in the experiment, (a)–(f) are used as reference images, respectively, and (g)–(i) are used as sensed images, respectively). (a) GE T1. (b) GE T2. (c) UAV OPT. (d) UAV SAR. (e) UAV DOM. (f) UAV DSM. (g) ZY3 PAN. (h) UAV IR. (i) GF3 SAR.

Fig. 5 displays thumbnail images, all of which are from the same geographic area. The images consist of data from Google Earth, Gaofen-3 satellite, Ziyuan-3 satellite, and UAV with different sensors. The spectral bands cover a range of optical, IR, and SAR wavelengths. Among them, GE T1, GE T2, UAV OPT, UAV SAR, UAV digital orthophoto map (DOM), and UAV digital surface model (DSM) are used as reference images, and ZY3 PAN, UAV IR, and GF3 SAR are used as sensed images. The purpose is to measure the proposed method's matching against optical, IR, and SAR images of the same scene.

*2) Qualitative Comparisons on the Same Scene Dataset:* We validated the matching performance of the proposed method using the same scene multimodal RSI dataset. Fig. 6 illustrates the qualitative results of our method compared to state-of-the-art matching methods on this dataset. The image pairs in the dataset pose significant challenges for matching. Specifically, as for reference images, Fig. 6(a) and (b) represents typical Google Earth images captured at different time periods, Fig. 6(c) and (d) shows optical and SAR images obtained from UAV with different imaging modes, and Fig. 6(e) and (f) depicts DOM and DSM generated from UAV images. The results demonstrate that the proposed method achieves a higher number of correct matches and fewer false matches compared to other state-of-the-art methods, even when matching with a DSM image as a reference. It is worth noting that although handcrafted- and learning-based methods exhibit excellent matching performance when applied to optical image pairs, they encounter challenges when dealing with IR and SAR images. In particular, there are notable differences in the number and distribution uniformity of matched keypoints, leading to unstable matching results. In contrast,

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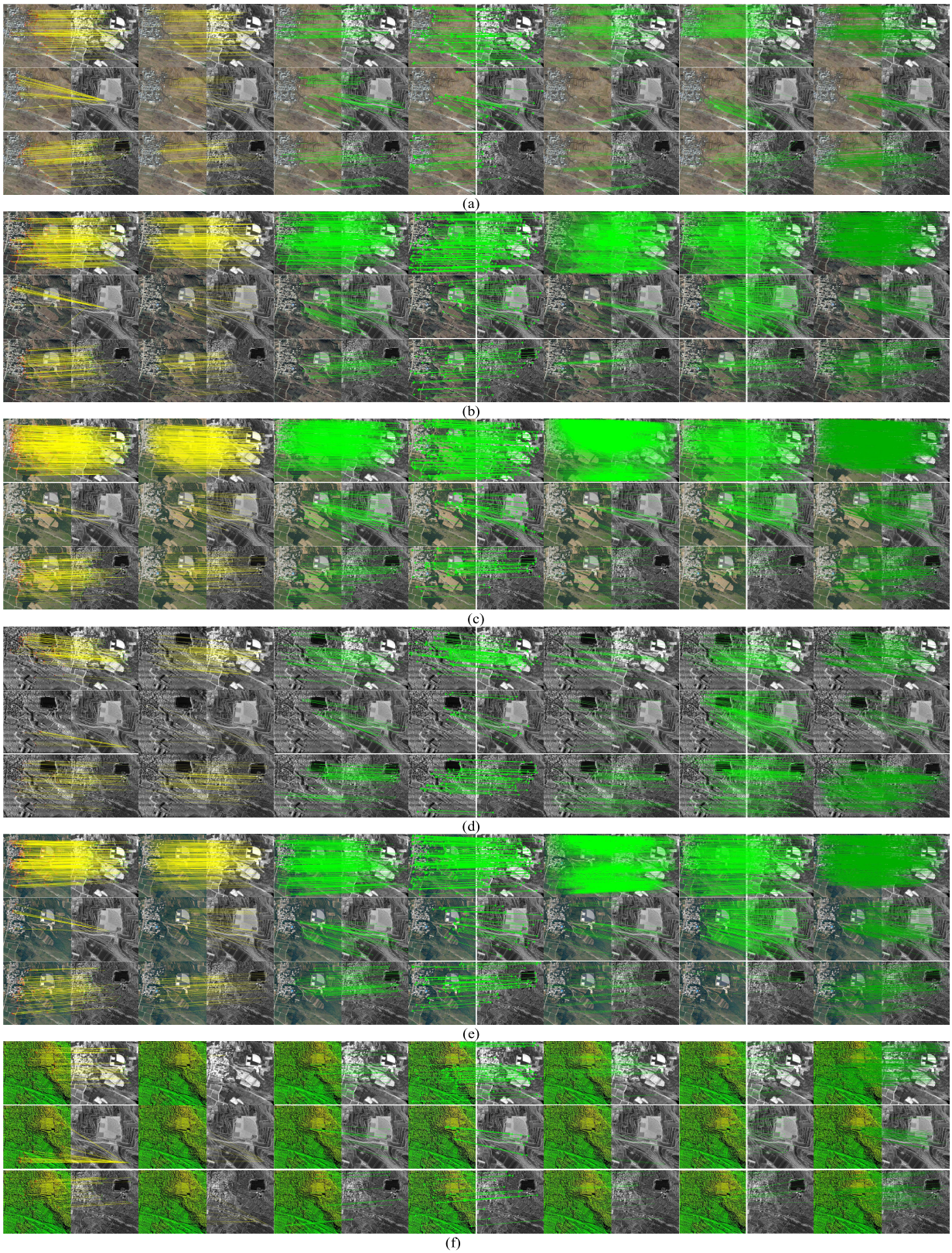


Fig. 6. Qualitative matching results (from (left) to (right): RIFT, ASS, D2Net, CMMNet, ECOTR, LightGlue, and MICM). (a) From (top) to (bottom): GE T1-ZY3 PAN, UAV IR, and GF3 SAR. (b) GE T2-ZY3 PAN, UAV IR, and GF3 SAR. (c) UAV OPT-ZY3 PAN, UAV IR, and GF3 SAR. (d) UAV SAR-ZY3 PAN, UAV IR, and GF3 SAR. (e) DOM-ZY3 PAN, UAV IR, and GF3 SAR. (f) DSM-ZY3 PAN, UAV IR, and GF3 SAR.

our method shows robust matching performance and modality invariance, effectively addressing the limitations observed in the other methods.

After analyzing the performance of other state-of-the-art methods on the same scene dataset, we have observed certain limitations in their matching capabilities. For example, RIFT is

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TABLE II
IMAGE MATCHING QUANTITATIVE RESUTLS ON THE SAME SCENE DATASET

| Reference | Sensed | RIFT | | ASS | | D2Net | | CMMNet | | ECOTR | | LightGlue | | MICM | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE |
| GE(T1) | ZY3(PAN) | 139 | 2.74 | 151 | 2.58 | 64 | 3.12 | 74 | 2.68 | 217 | 1.78 | 229 | 1.78 | **336** | **1.64** |
| | UAV(IR) | 0 | -- | 27 | 2.55 | 54 | 3.87 | 20 | 5.25 | 12 | -- | 54 | 4.83 | **177** | **2.38** |
| | GF3(SAR) | 90 | 3.68 | 60 | 3.96 | 33 | 4.55 | 21 | -- | 34 | 5.96 | 0 | -- | **232** | **1.76** |
| GE(T2) | ZY3(PAN) | 323 | 1.77 | 454 | 1.6 | 466 | 1.16 | 197 | 1.52 | **2219** | **0.82** | 741 | 1.36 | 1043 | 1.33 |
| | UAV(IR) | 0 | -- | 24 | 2.76 | 60 | 3.77 | 30 | 4.03 | 37 | 4.77 | **396** | **1.55** | 315 | 1.87 |
| | GF3(SAR) | 93 | 3.36 | 72 | 3.85 | 40 | 4.58 | 41 | 3.96 | 47 | 5.36 | 85 | 4.85 | **312** | **1.43** |
| UAV(OPT) | ZY3(PAN) | 741 | 0.6 | 1010 | 0.55 | 738 | 1.82 | 250 | 1.35 | **4493** | **0.65** | 931 | 1.03 | 1992 | 0.92 |
| | UAV(IR) | 0 | -- | 50 | 2.13 | 88 | 2.73 | 36 | 3.34 | 38 | -- | 233 | 2.06 | **520** | **1.68** |
| | GF3(SAR) | 151 | 2.76 | 107 | 2.88 | 51 | 3.14 | 54 | 2.89 | 32 | 5.36 | 26 | -- | **336** | **1.39** |
| UAV(SAR) | ZY3(PAN) | 109 | 3.15 | 97 | 3.28 | 56 | 3.52 | 70 | 2.88 | 21 | 4.38 | 130 | 3.14 | **292** | **2.15** |
| | UAV(IR) | 0 | -- | 6 | -- | 23 | 5.18 | 12 | -- | 12 | -- | **234** | **2.22** | 170 | 2.67 |
| | GF3(SAR) | 135 | 3.36 | 64 | 3.86 | 66 | 3.22 | 45 | 3.46 | 91 | 3.76 | 259 | 2.17 | **521** | **1.88** |
| DOM | ZY3(PAN) | 455 | 1.36 | 623 | 1.22 | 418 | 1.33 | 199 | 2.32 | **2773** | **0.63** | 815 | 0.90 | 1197 | 0.87 |
| | UAV(IR) | 0 | -- | 42 | 2.15 | 54 | 2.94 | 25 | 4.36 | 35 | 5.03 | 373 | 1.52 | **481** | **1.34** |
| | GF3(SAR) | 101 | 2.68 | 77 | 3.52 | 57 | 3.66 | 49 | 3.73 | 38 | 4.66 | 0 | -- | **278** | **1.22** |
| DSM | ZY3(PAN) | 51 | 2.46 | 0 | -- | 32 | 4.34 | 54 | 3.56 | 59 | 4.35 | 0 | -- | **312** | **2.85** |
| | UAV(IR) | 0 | -- | 0 | -- | 11 | 5.76 | 11 | -- | 14 | -- | 0 | -- | **190** | **3.52** |
| | GF3(SAR) | 29 | 3.28 | 0 | -- | 13 | 6.11 | 20 | 5.15 | 14 | -- | 0 | -- | **178** | **2.96** |

sensitive to scale and speckle noise differences. This sensitivity may arise from inaccurate edge structure information in the phase congruency map caused by speckle noise, as discussed by Li et al. [59]. Although ASS exhibits better adaptability on this dataset, it still yields fewer correct matches on SAR and IR images. This can be attributed to the difficulty in extracting higher levels of semantic information from these modalities [27]. Similarly, the four learning-based methods display robustness in handling optical images but demonstrate sensitivity toward IR and SAR images. This sensitivity can be attributed to two main factors: 1) the training dataset lacks multimodal RSIs and 2) the models' generalization performance is not sufficiently robust. Furthermore, all compared methods exhibit sensitivity to DSM images, which results in a limited range of matching adaptability.

*3) Quantitative Comparisons on the Same Scene Dataset:* Table II lists the quantitative matching results of the MICM and comparative methods in terms of CM and RMSE, which are vital metrics. The results indicate that the MICM method achieves a high CM and a small RMSE. It shows that the proposed hybrid local multiscale learning features and transformer attention mechanism yield superior performance in multimodal RSI matching. Specifically, the average CM value is 493, while the average RMSE value is 1.88. These values further show the suitability for matching multimodal RSI with nonlinear geometric and radiometric distortions.

Among the other state-of-the-art methods, RIFT and ASS represent comparable matching performance in optical and SAR image matching. However, ASS outperforms RIFT on IR images, potentially due to its utilization of local statistics weighted difference operation to suppress image noises. However, for the DSM image, it is not effective, leading to performance degradation. When DSM is used as a reference, all comparative methods struggle to fit complex characteristics, resulting in decreased performance. Regarding the learning-based methods, comparative methods exhibit comparable performance. This can be attributed to the generation of a receptive field by the multilayer convolution of CNN, which extracts more stable features and yields distinguishable feature descriptors. These learning-based methods demonstrate excellent CM and RMSE results for optical sensed images but vary in performance for IR and SAR images. Notably, the compared methods are note suitable for matching UAV SAR–UAV IR images and DSM as reference images. In addition, the difference in single-scale and multiscale D2Net lies in the stability of the latter, indicating that the extraction of multiscale features significantly enhances the adaptability of multimodal RSI matching in the presence of nonlinear radiometric differences.

To sum up, the proposed MICM method outperformed the comparative methods in terms of CM and RMSE, particularly for IR and SAR sensed images, and DSM was used as the reference image. This indicates that our method exhibits better matching adaptability on the same scene dataset. Visualization of qualitative matching results shows that our method has modalities invariant, validating the usefulness of the hybrid CNN feature and transformer attention mechanism. The key reason may be threefold: 1) we fine-tuned the backbone network model to leverage the nonlinear fitting ability of a deep CNN; 2) we combined local and global features to accurately identify high-quality correspondence keypoints in multimodal RSIs; and 3) we designed a loss function to encourage the exploration of more corresponding keypoints, making our method more conducive to multimodal RSIs' matching.

Fig. 7 shows a comparative analysis of the coarse matching (NIM) and matching time (MT) for the same scene dataset. Examining the 25%–75% interquartile range (IQR), the range

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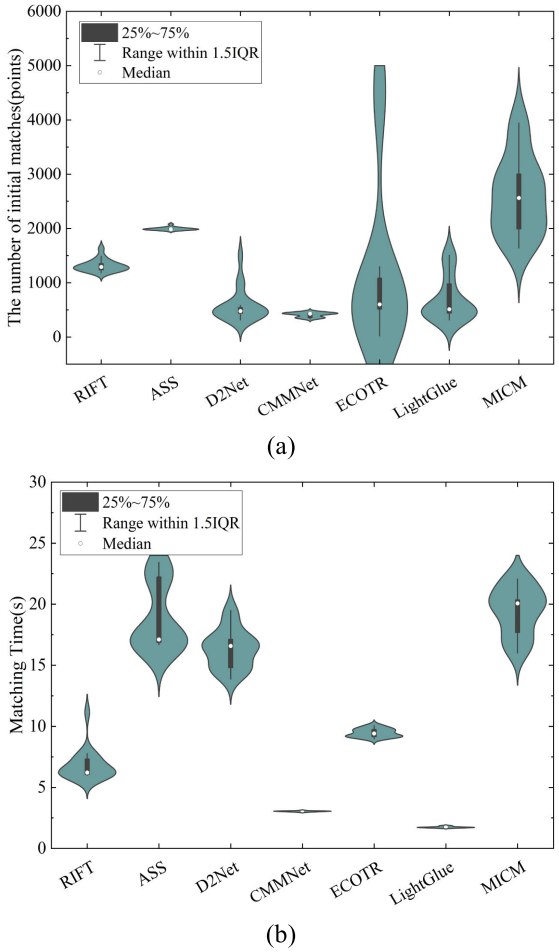


Fig. 7. Performance comparison of different methods on the same scene dataset. (a) NIM. (b) MT.

within 1.5 times the IQR (1.5 IQR), and the median, it shows that the proposed method achieves the highest median NIM, maintaining a 25%–75% IQR between 2000 and 3000 pairs. This provides favorable initial conditions for obtaining rich and accurate matching results. Among the comparative methods, ASS exhibits the most stable NIM, consistently around 2000, while RIFT demonstrates a relatively stable NIM, distributed around 1500. Comparing learning-based matching methods with handcrafted matching methods reveals that the latter exhibits greater stability in terms of NIM. In terms of MT, the comparative methods are generally slightly better than MICM. Particularly, LightGlue and CMMNet methods exhibit higher matching efficiency, significantly outperforming the MT of MICM. This is mainly attributed to the proposed method's use of a transformer attention mechanism based on an encoder–decoder structure. The decoder's regression of the initial positions of keypoints consumes more runtime, resulting in higher MT. However, considering the outstanding matching performance achieved by our method, for applications where MT requirements are not stringent, spending a small amount of time to obtain better results is an acceptable tradeoff.

*4) Performance of Image Registration:* The proposed method not only demonstrates effectiveness in feature matching but also extends its application to image registration, which is an important task in multimodal RSI analysis. In the registration task, the focus is on maximizing the alignment of the overlapping areas between the sensed image and the reference image. Fig. 8 showcases the results in a chessboard mosaic, with rectangles indicating enlarged details. The checkerboard diagrams clearly demonstrate that the proposed method achieves satisfactory registration effects for all the experimental data. It effectively adapts to the nonlinear radiation and geometric differences present in the data, including variations in time, scale, season, texture, imaging mode, and imaging sensors. This demonstrates the versatility and robustness of the proposed method in handling multimodal RSI registration tasks.

The effectiveness in obtaining evenly distributed and sufficient matches on optical, IR, and SAR images plays a crucial role in accurately calculating the image transformation parameters. This capability ensures that the proposed method can provide reliable and precise matching and registration results for multimodal RSIs. By considering the unique characteristics and challenges presented by different imaging modes, our method demonstrates its ability to handle diverse modalities in multimodal RSI matching and registration tasks.

### *E. Experiments on the Different Scene Dataset*

*1) Description of the Different Scene Dataset:* The dataset comprises images obtained from diverse ground surfaces using different imaging modes and platforms at different locations, as indicated in Table III. Detailed information such as the image data source, band type, pixel size, and resolution is provided for each image pair in the dataset. The images were captured using a range of sensors, incorporating variations in rotations, times, resolutions, scales, textures, and so on. The data sources are from Google Earth, satellites, and UAV platforms, making it highly suitable for evaluating the performance of the proposed algorithm in handling diverse data modalities. Fig. 9 presents a thumbnail view of the experimental images, all of which are captured from different scenes. The primary objective is to assess the generalization capability of the proposed method when applied to optical, IR, and SAR images obtained from different scene areas. Moreover, this dataset contains a wide variety of data types, landforms, ground objects, scales, and times, providing a comprehensive evaluation platform for assessing the matching performance of the proposed method in terms of matching accuracy and efficiency.

*2) Qualitative Comparisons on the Different Scene Dataset:* After presenting the challenging dataset for multimodal RSI matching in various scenarios, Fig. 10 illustrates the visualization matching results on the different scene dataset. It is evident that the proposed method successfully identifies numerous correct feature keypoints in all the tested image pairs, and the matched keypoints are evenly distributed across the multimodal RSIs. Notably, our method exhibits a high number of matches in DSM-GF3 SAR and DSM-ZY3 PAN image pairs, indicating its scalability and modality invariance. In contrast, other SOTA methods exhibit limitations, particularly in handling rotation, scale, texture, viewpoint, or imaging mechanism differences. Overall, the proposed MICM method

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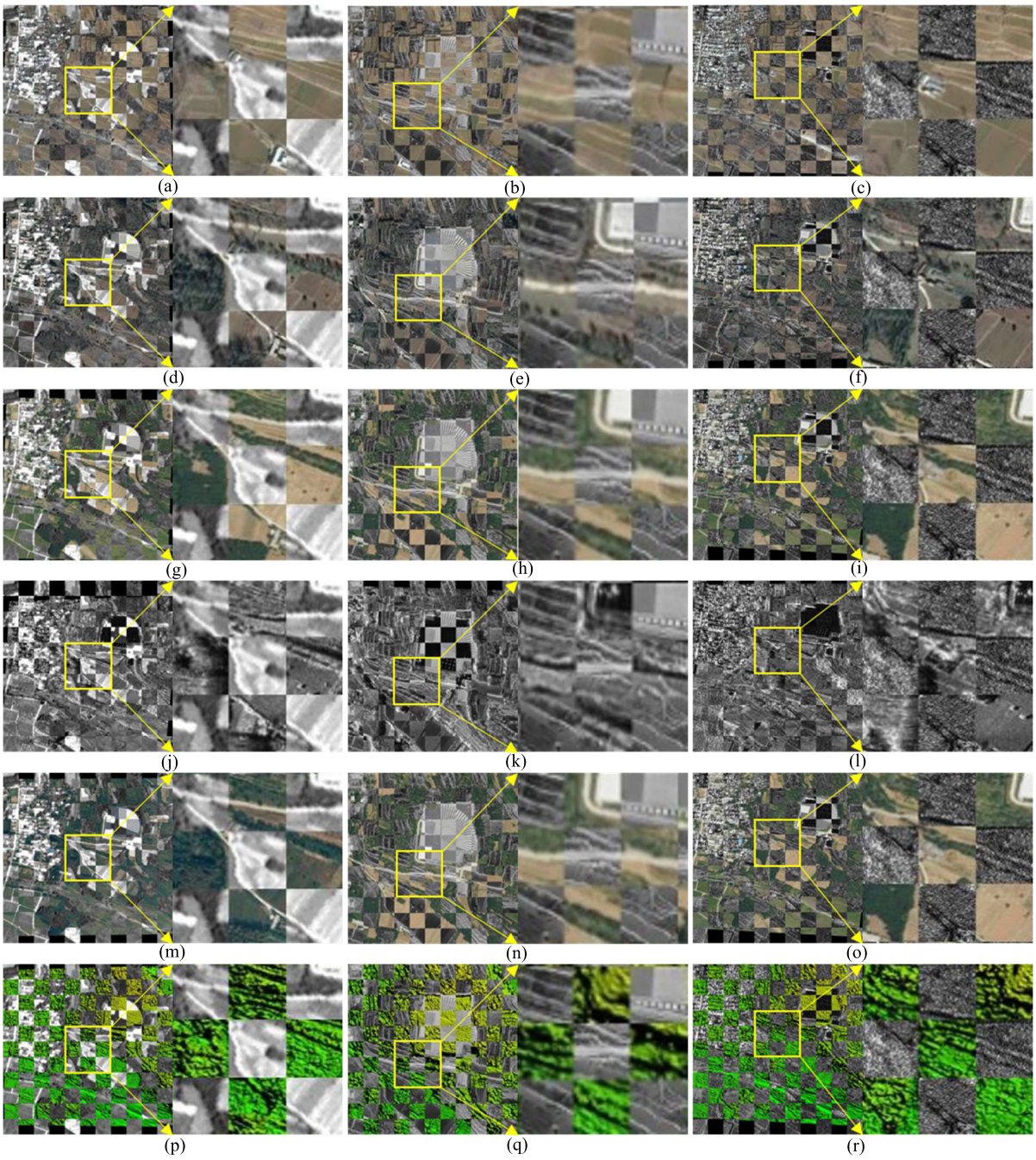


Fig. 8. Qualitative results of multimodal RSI registration. (a) GE T1-ZY3 PAN. (b) GE T1-UAV IR. (c) GE T1-GF3 SAR. (d) GE T2-ZY3 PAN. (e) GE T2-UAV IR. (f) GE T2-GF3 SAR. (g) UAV OPT-ZY3 PAN. (h) UAV OPT-UAV IR. (i) UAV OPT-GF3 SAR. (j) UAV SAR-ZY3 PAN. (k) UAV SAR-UAV IR. (l) UAV SAR-GF3 SAR. (m) UAV DOM-ZY3 PAN. (n) UAV DOM-UAV IR. (o) UAV DOM-GF3 SAR. (p) UAV DSM-ZY3 PAN. (q) UAV DSM-UAV IR. (r) UAV DSM-GF3 SAR.

demonstrates excellent stability in multimodal RSI matching, achieving a substantial number of correct matches across variations in scale, viewpoint, rotation, and time. Therefore, the proposed method serves as a valuable reference for multimodal RSIs' matching challenges.

*3) Quantitative Comparisons on the Different Scene Dataset:* Table IV provides the quantitative matching results of our method and the compared methods in terms of CM and RMSE. Although all methods exhibit improved results on the different scene dataset compared to the same scene dataset, it can be seen that the MICM method maintains a significant performance advantage. It outperforms the compared methods in terms of CM and RMSE, reaffirming its superiority in handling diverse data modalities and addressing nonlinear geometric and radiometric distortions in multimodal RSIs. The average CM and RMSE values across the entire dataset are 924 and 1.46, respectively, and the average CM and RMSE values for the multisource Google Earth images, satellite images, and UAV images are (1224, 1.62), (633, 1.25), and (915, 1.51). This performance lead highlights the effectiveness and robustness of the proposed method in achieving accurate and reliable multimodal RSI matching across various scenes.

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TABLE III
EXPERIMENTAL DATA DESCRIPTION BASED ON THE DIFFERENT SCENES

| Platform | No. | Sensor | Pixel | Resolution(m) | Characteristic |
|---|---|---|---|---|---|
| Google Earth | 1 | OPT-OPT | 512×512 / 512×512 | 2.0 / 2.0 | These images are all from Google earth platform and have significant differences in times, rotations, illumination, scales, and types of ground objects. The terrain is mainly plain area. The scene area contains urban, suburban, and villages. They are used to test the proposed algorithm's performance for google earth images. |
| | 2 | OPT-OPT | 512×512 / 512×512 | 1.0 / 1.0 | |
| | 3 | OPT-OPT | 512×512 / 512×512 | 2.0 / 2.0 | |
| | 4 | OPT-OPT | 512×512 / 512×512 | 5.0 / 5.0 | |
| | 5 | OPT-OPT | 512×512 / 512×512 | 1.0 / 1.0 | |
| | 6 | OPT-OPT | 512×512 / 512×512 | 1.0 / 1.0 | |
| Satellite | 7 | OPT-SAR | 512×512 / 512×512 | 1.0 / 2.0 | These images come from satellite platforms with different imaging sensors. There are obvious differences in scale, texture and imaging mode. The terrain includes plain, mountainous region. They are mainly used to test the proposed method's performance in terms of matching multimodal satellite images. |
| | 8 | OPT-SAR | 512×512 / 512×512 | 1.0 / 1.0 | |
| | 9 | OPT-SAR | 512×512 / 512×512 | 2.0 / 5.0 | |
| | 10 | OPT-OPT | 512×512 / 512×512 | 5.0 / 5.0 | |
| | 11 | OPT-OPT | 512×512 / 512×512 | 30.0 / 30.0 | |
| | 12 | OPT-OPT | 512×512 / 512×512 | 1.0 / 2.0 | |
| UAV | 13 | OPT-IR | 512×512 / 512×512 | 0.05 / 0.1 | The images of these different modes come from the UAV platform, which suffer from significant differences in time, scale, viewpoint, and imaging mechanism. The scene areas are mainly suburban and villages. They are used to test the performance of the proposed algorithm in terms of multimodal UAV image matching. |
| | 14 | OPT-IR | 512×512 / 512×512 | 0.05 / 0.1 | |
| | 15 | OPT-IR | 512×512 / 512×512 | 0.05 / 0.1 | |
| | 16 | SAR-SAR | 512×512 / 512×512 | 0.5 / 0.5 | |
| | 17 | OPT-OPT | 512×512 / 512×512 | 0.1 / 0.05 | |
| | 18 | OPT-OPT | 512×512 / 512×512 | 0.1 / 0.1 | |

Among the six state-of-the-art methods used for comparison, there are excellent matching performances in many scenes; however, there are some incorrect matches on the compared methods in multisource satellite image pairs and UAV image pairs. In these image pairs, significant differences in scale, viewpoint, and rotation are presented, highlighting the robustness of the proposed MICM method in handling such challenging scenarios. In addition, compared to other methods, our method achieves smaller and more stable RMSE results, indicating its ability to achieve stable multimodal RSI high-precision matching.

Fig. 11 displays the quantitative results of the proposed method and comparative methods in terms of coarse matching quantity (NIM) and MT metrics. Examining the 25%–75% IQR, the range within 1.5 IQR, and the median, it is evident that the proposed method exhibits higher NIM values for the NIM metric. Although there is a considerable difference between the maximum and minimum NIM values, both values indicate abundant feature-matching points, demonstrating the ability of the proposed method to handle various challenging scenarios. Regarding the MT, comparative methods are slightly faster than the proposed method. This is related to the transformer attention mechanism with rich feature keypoints. However, considering the improvement of the matching performance for the proposed method, it is acceptable to sacrifice a little running time.

## V. DISCUSSION

In this section, we will perform several detailed ablation experiments of different backbone networks, different NMS window sizes, and image noises on the number of correct matches to discuss the efficiency and effectiveness of the proposed MICM method.

### A. Performance Comparison Between Different Backbone Network Architectures

In order to design a suitable backbone network that can efficiently operate on multimodal RSI and preserve high- and low-level feature information in as much detail as possible, we investigate two classical backbone networks: ResNet50 and our proposed Unet-like backbone. For the ResNet50 network, we cropped it at layer 3 (before the ReLU unit) and followed by the feature L2-Norm. From Fig. 12, we observed that the Unet-like backbone outperformed the ResNet50 backbone model, showing higher correspondence matching scores on both the same scene and different scene datasets. This indicates that the proposed Unet-like backbone can extract more correspondence feature keypoints in multimodal RSI. When using the Unet-like backbone network for multimodal RSI matching, we achieved RMSE within 4 pixels, with most of them falling within 3 pixels. In contrast, the ResNet50-based backbone exhibited large fluctuations and failed to match two locations, as shown in the magenta regions in Fig. 12(c). This highlights the effectiveness of the Unet-like backbone network in matching multimodal RSIs. The key reason is that the Unet-like backbone can extract multiscale invariant features, and the NMS sampling strategy helps to obtain feature points with a uniform distribution.

### B. Performance Comparison Between Different NMS Size

To examine the impact of the NMS window size, we conducted experiments on the same scene dataset to evaluate the multimodal RSI matching results. The NMS window sizes considered in our experiment were $1 \times 1$, $3 \times 3$, $5 \times 5$, $7 \times 7$, $9 \times 9$, and $11 \times 11$. Fig. 13 demonstrates the number of feature extraction and matching runtime for different window sizes. We observed that the $1 \times 1$ NMS window size

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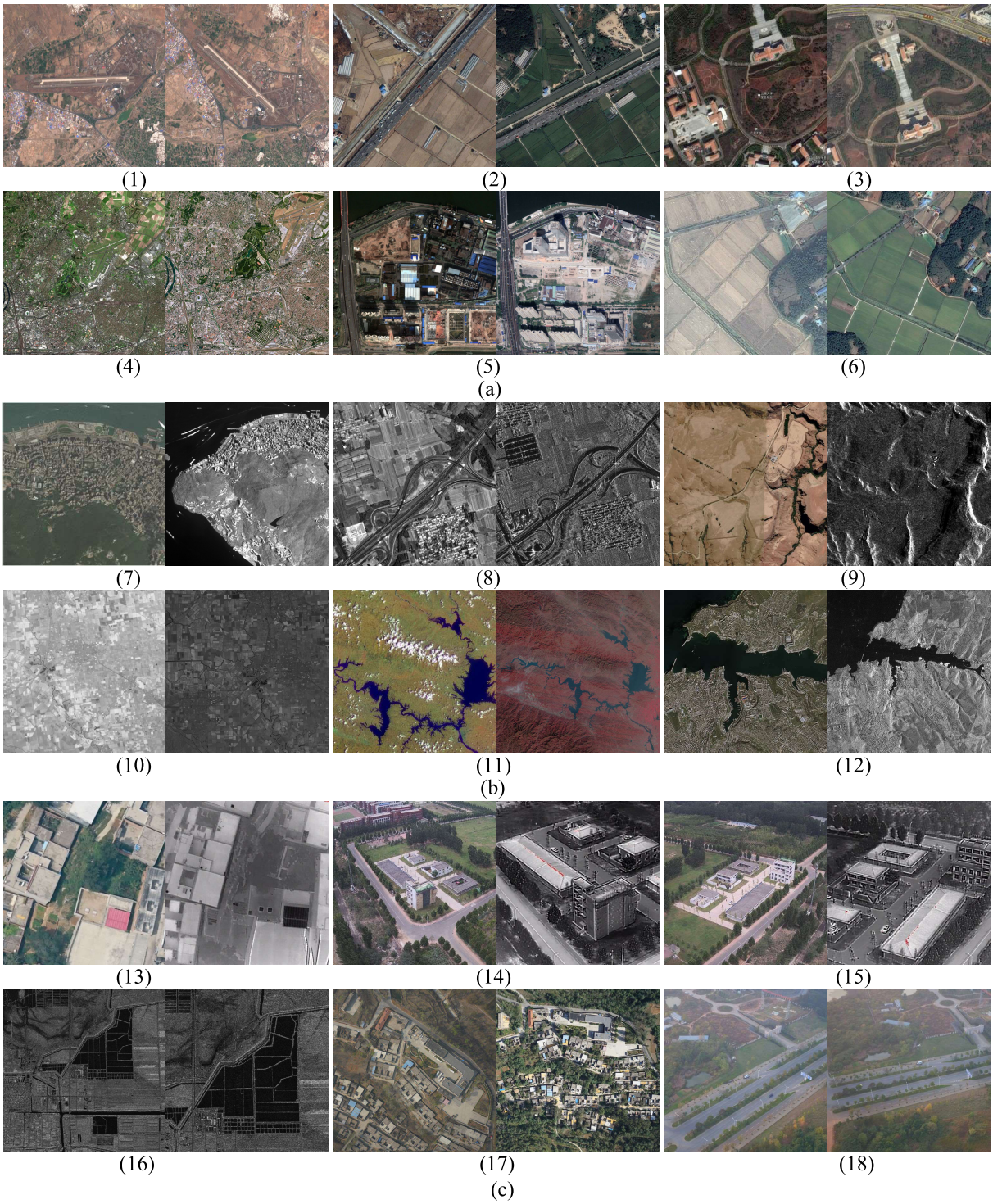


Fig. 9. Multimodal RSIs of the different scene dataset. (a) Multisource Google Earth images. (b) Multisource satellite images. (c) Multisource UAV images.

resulted in the highest number of feature keypoints but also the longest runtime, exceeding the number of feature detections and runtimes of other window sizes. In the enlarged local detail window, we can see that the $3 \times 3$ window size yielded a more abundant number of features, with a similar MT compared to other window sizes. As a result, we selected the $3 \times 3$ window size as the final choice for the NMS window size.

In the following analysis, we explore the impact of different NMS window sizes on the number of CMs and the RMSE, as presented in Table V. Several conclusions are reached from the results.

1) The $1 \times 1$ window size of the NMS achieves the highest number of correct matches and the lowest RMSE. As the NMS window size decreases, the number of fine feature matches increases, and the RMSE value decreases. This indicates that the NMS window size has a significant effect on the quality of multimodal RSI matching.

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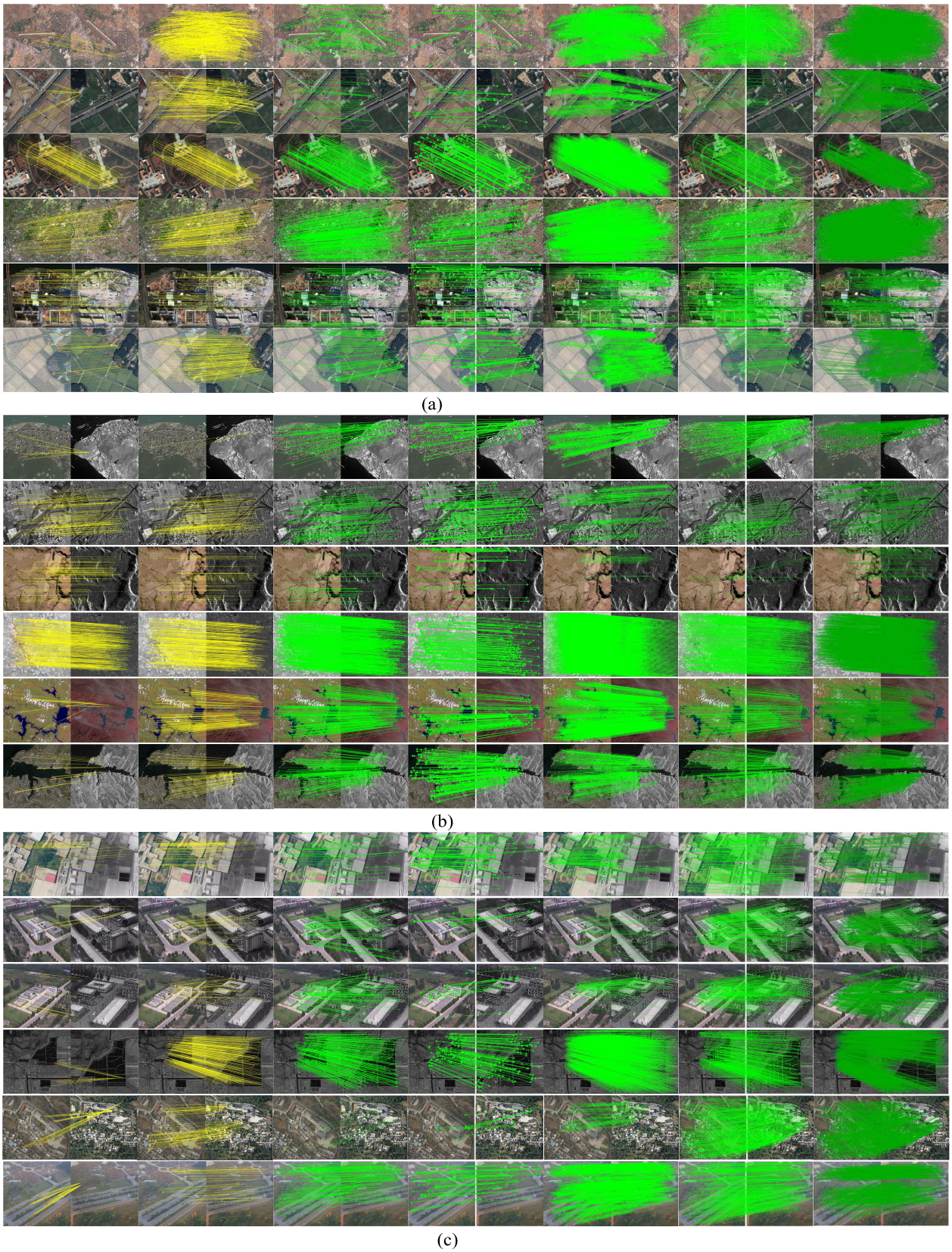

Fig. 10. MMRS image-matching results of the different scene (RIFT, ASS, D2Net, CMMNet, ECOTR, LightGlue, and MICM). (a) Multisource Google Earth image-matching results. (b) Multisource satellite image-matching results. (c) Multisource UAV image-matching results.

2) Table V can be divided into three categories based on the different NMS window sizes. The first category corresponds to the CM and RMSE results for the $1 \times 1$ window size, which demonstrates pixel-level matching processing. Although it achieves excellent image-matching performance, it may not adapt well to

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TABLE IV
IMAGE MATCHING QUANTITATIVE RESUTLS ON THE SAME SCENE DATASET

| Types | Pairs | RIFT | | ASS | | D2Net | | CMMNet | | ECOTR | | LightGlue | | MICM | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE |
| Multisource Google Earth images | Pair1 | 80 | -- | 1163 | **1.28** | 110 | 3.05 | 21 | -- | 1347 | 1.36 | 1220 | 1.63 | **2124** | 1.52 |
| | Pair2 | 272 | 2.74 | 336 | 2.12 | 37 | 5.12 | 20 | 5.26 | 309 | 2.44 | 49 | 4.84 | **817** | **1.88** |
| | Pair3 | 207 | 1.58 | 247 | 1.03 | 266 | 1.00 | 155 | 1.13 | **1883** | **0.53** | 327 | 1.01 | 635 | 0.97 |
| | Pair4 | 42 | 1.26 | 270 | 0.95 | 427 | 0.88 | 76 | 1.15 | 1769 | 0.51 | 502 | 0.88 | **2348** | **0.64** |
| | Pair5 | 100 | 3.87 | 91 | 3.55 | 74 | 3.67 | 83 | 3.52 | 748 | **2.66** | 619 | 2.35 | **765** | 2.86 |
| | Pair6 | 44 | 5.52 | 255 | 2.01 | 101 | 3.23 | 62 | 4.28 | 699 | 2.15 | 118 | 3.55 | **653** | **1.88** |
| Multisource Satellite images | Pair1 | 11 | -- | -- | -- | 74 | 2.66 | 50 | 2.99 | **332** | **1.64** | 338 | 1.74 | 250 | 1.95 |
| | Pair2 | 180 | 1.66 | 164 | 1.56 | 116 | 1.65 | 92 | 1.96 | 183 | 1.77 | 128 | 2.62 | **508** | **1.23** |
| | Pair3 | 122 | 1.62 | 70 | 1.85 | 26 | 4.96 | 33 | 3.36 | 38 | -- | 15 | -- | **234** | **1.67** |
| | Pair4 | 566 | 0.60 | 628 | 0.51 | 972 | 0.73 | 151 | 0.83 | **3426** | **0.48** | 938 | 0.64 | 1604 | 0.56 |
| | Pair5 | 0 | -- | 242 | 0.78 | 300 | 1.12 | 127 | 1.05 | **1737** | **0.85** | 321 | 0.97 | 586 | 0.87 |
| | Pair6 | 36 | 4.31 | 100 | 1.54 | 199 | 1.87 | 205 | 1.11 | 534 | 1.16 | 305 | 1.53 | **619** | **1.22** |
| Multisource UAV images | Pair1 | 43 | 3.87 | 94 | 2.36 | 62 | 2.89 | 69 | 2.67 | 221 | 2.25 | 482 | 1.83 | **581** | **1.64** |
| | Pair2 | 0 | -- | 37 | 4.41 | 62 | 2.64 | 11 | -- | 42 | 4.87 | 337 | 2.36 | **608** | **1.76** |
| | Pair3 | 0 | -- | 44 | 3.36 | 97 | 3.16 | 13 | -- | 97 | 4.32 | 312 | 2.59 | **758** | **2.34** |
| | Pair4 | 19 | 4.88 | 440 | 0.80 | 319 | 1.54 | 91 | 1.66 | 1782 | 0.99 | 602 | 0.93 | **1362** | **0.85** |
| | Pair5 | 62 | 3.56 | 117 | 1.66 | 10 | -- | 9 | -- | 85 | 5.36 | 706 | 1.55 | **857** | **1.35** |
| | Pair6 | 10 | -- | 61 | 1.95 | 178 | 2.01 | 57 | 3.27 | 1138 | 1.15 | 705 | 1.32 | **1327** | **1.13** |

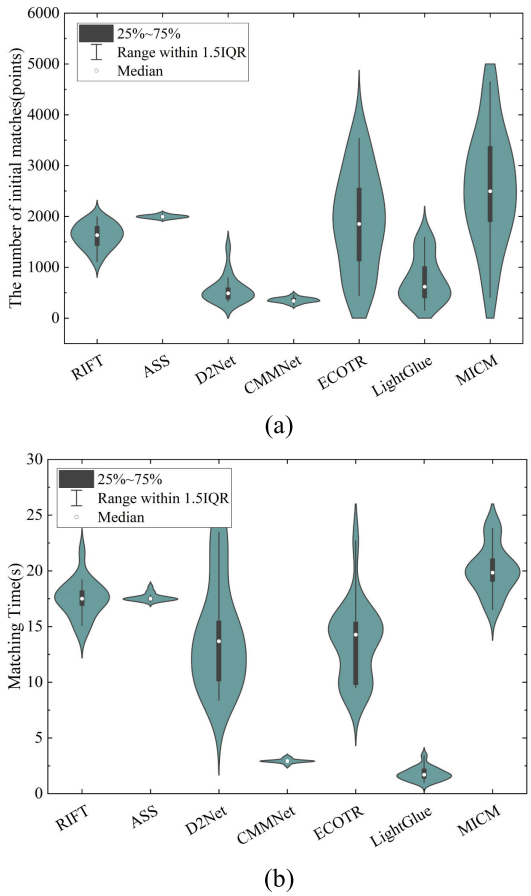


Fig. 11. Performance comparison of different methods on the different scene dataset. (a) NIM. (b) MT.

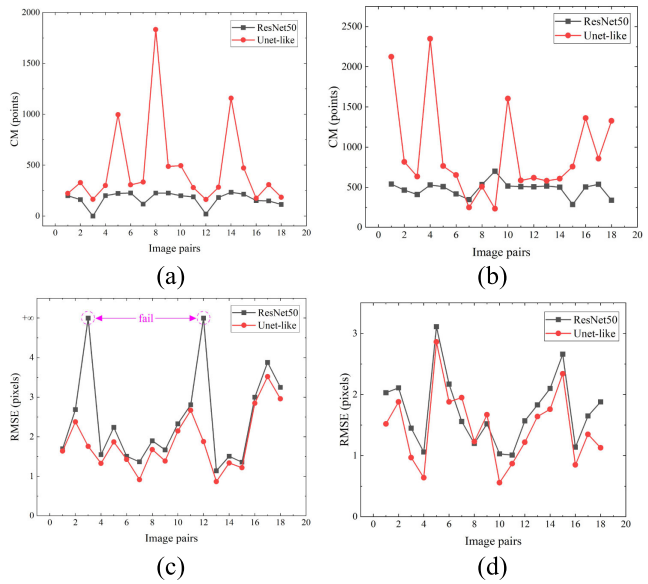


Fig. 12. Performance comparison of different backbones. (a) Results of CM on the same scene dataset. (b) Results of CM on the different scene dataset. (c) Results of RMSE on the same scene dataset. (d) Results of RMSE on the different scene dataset.

sparse feature matching. The second category includes the CM and RMSE results for the $3 \times 3$, $5 \times 5$, and $7 \times 7$ window sizes, which exhibits similar CM and RMSE values. The third category comprises the results for other NMS window sizes, which are more sensitive to multimodal RSIs.

3) The average values of CM and RMSE show that our method performs well in multimodal RSI matching. Through the above analysis, we determine that the optimal NMS window size is $3 \times 3$. This is because this setting yields higher CM and smaller RMSE values, striking a balance between matching accuracy and efficiency. This further indicates that our NMS detection can robustly handle multimodal RSI matching tasks.

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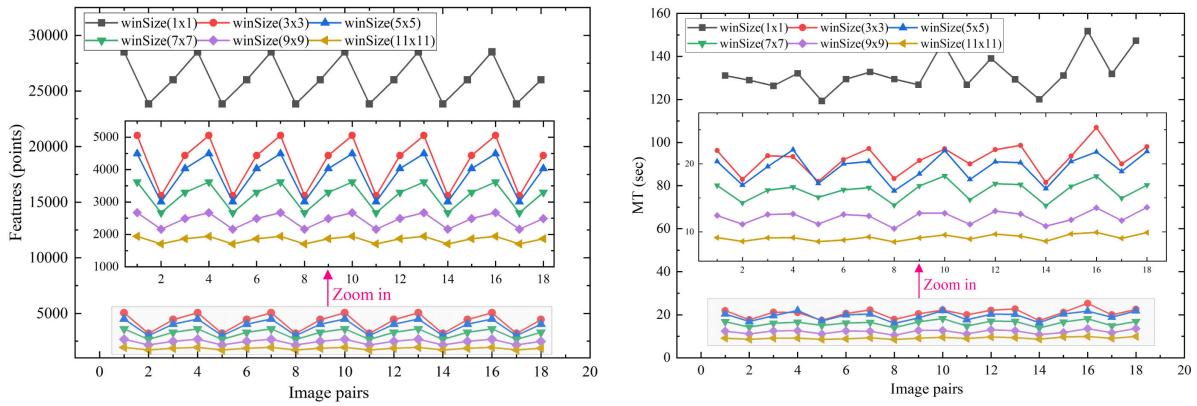


Fig. 13. Performance comparison on the same scene dataset in terms of different NMS window sizes. (a) Features. (b) MT.

TABLE V
IMAGE MATCHING QUANTITATIVE RESUTLS ON THE SAME SCENE DATASET

| Reference | Sensed | winSize (1×1) | | winSize (3×3) | | winSize (5×5) | | winSize (7×7) | | winSize (9×9) | | winSize (11×11) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE | CM | RMSE |
| GE(T1) | ZY3(PAN) | 485 | 1.55 | 336 | 1.64 | 329 | 1.66 | 317 | 1.67 | 297 | 1.75 | 258 | 1.81 |
| | UAV(IR) | 226 | 2.18 | 177 | 2.38 | 165 | 2.39 | 160 | 2.40 | 146 | 2.53 | 136 | 2.62 |
| | GF3(SAR) | 295 | 1.71 | 232 | 1.76 | 223 | 1.80 | 214 | 1.82 | 179 | 1.93 | 162 | 1.97 |
| GE(T2) | ZY3(PAN) | 1758 | 1.21 | 1043 | 1.33 | 997 | 1.41 | 926 | 1.43 | 818 | 1.54 | 708 | 1.61 |
| | UAV(IR) | 419 | 1.66 | 315 | 1.87 | 307 | 1.90 | 289 | 1.91 | 250 | 1.99 | 211 | 2.09 |
| | GF3(SAR) | 349 | 1.40 | 312 | 1.43 | 300 | 1.45 | 290 | 1.45 | 266 | 1.66 | 230 | 1.74 |
| UAV(OPT) | ZY3(PAN) | 4550 | 0.65 | 1992 | 0.92 | 1833 | 0.91 | 1608 | 0.92 | 1320 | 1.05 | 1044 | 1.07 |
| | UAV(IR) | 764 | 1.47 | 520 | 1.68 | 488 | 1.70 | 460 | 1.71 | 383 | 1.82 | 304 | 1.88 |
| | GF3(SAR) | 391 | 1.28 | 336 | 1.39 | 335 | 1.36 | 315 | 1.37 | 272 | 1.49 | 234 | 1.65 |
| UAV(SAR) | ZY3(PAN) | 324 | 2.07 | 292 | 2.15 | 280 | 2.21 | 279 | 2.20 | 244 | 2.34 | 197 | 2.37 |
| | UAV(IR) | 217 | 2.53 | 170 | 2.67 | 163 | 2.68 | 152 | 2.69 | 140 | 2.75 | 130 | 2.82 |
| | GF3(SAR) | 630 | 1.75 | 521 | 1.88 | 495 | 1.92 | 455 | 1.95 | 385 | 2.03 | 318 | 2.16 |
| DOM | ZY3(PAN) | 2109 | 0.63 | 1197 | 0.87 | 1158 | 0.85 | 1053 | 0.88 | 898 | 0.97 | 758 | 0.98 |
| | UAV(IR) | 739 | 1.12 | 481 | 1.34 | 473 | 1.33 | 444 | 1.35 | 371 | 1.46 | 315 | 1.58 |
| | GF3(SAR) | 397 | 1.15 | 278 | 1.22 | 283 | 1.20 | 256 | 1.23 | 247 | 1.28 | 220 | 1.30 |
| DSM | ZY3(PAN) | 389 | 2.69 | 312 | 2.85 | 308 | 2.85 | 298 | 2.88 | 273 | 2.97 | 250 | 2.99 |
| | UAV(IR) | 259 | 3.14 | 190 | 3.52 | 185 | 3.55 | 181 | 3.57 | 163 | 3.83 | 143 | 3.85 |
| | GF3(SAR) | 169 | 2.87 | 178 | 2.96 | 175 | 2.98 | 154 | 3.11 | 122 | 3.54 | 113 | 3.57 |
| Average | | 804 | 1.73 | 493 | 1.88 | 472 | 1.90 | 436 | 1.92 | 376 | 2.05 | 318 | 2.11 |

### C. Analysis of Noise Sensitivity

To assess the antinoise performance of the proposed method, we conducted evaluations by introducing different levels of Gaussian noise and speckle noise to the same scene images. Specifically, two sets of noisy images were generated: one with Gaussian noise and the other with speckle noise. For the Gaussian noise evaluation, we added different levels of Gaussian noise with mean 0 and variance $v$, where $v$ varied in the range [0, 0.5] with an interval of 0.1. This resulted in a series of noisy images with increasing levels of Gaussian noise. For the speckle noise evaluation, we added different levels of speckle noise with variances $v$, where $v$ varied in the range [0, 0.5] with an interval of 0.1. This produced a series of noisy images with increasing levels of speckle noise.

Fig. 14 shows the CM results of different levels of Gaussian noise and speckle noise on the same scene dataset. As the variance $v$ increases, the CM value exhibits a decreasing trend, indicating that the variance $v$ has a significant impact on the CM results. The proposed method shows adaptability to Gaussian noise up to a variance $v$ value of 0.3 and can handle speckle noise up to a variance $v$ value of 0.5. In practical applications, the variance value of Gaussian noise typically falls within the range of 0.01, while the variance value of speckle noise is usually within the range of 0.1. Therefore, the proposed method demonstrates high robustness in the presence of Gaussian and speckle noises.

The above results and coherence analysis highlight the clear effectiveness and advantages of the proposed method in handing significant nonlinear radiation differences and noise

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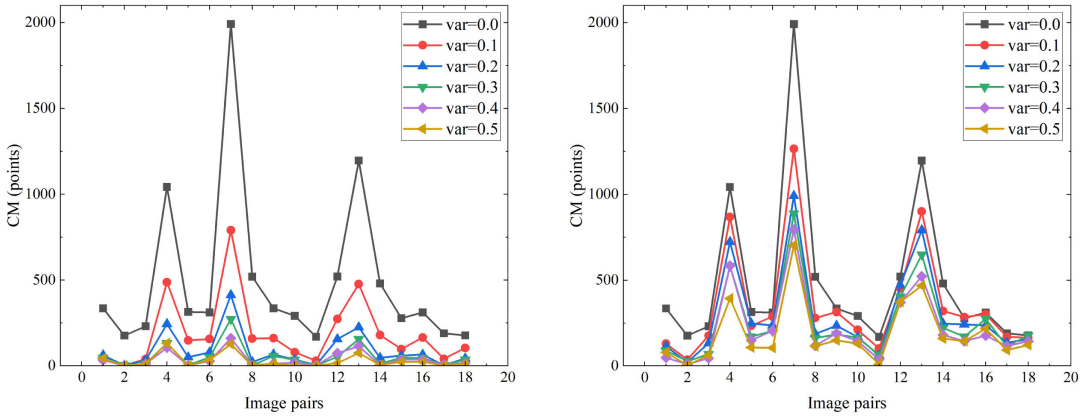


Fig. 14. CM results of Gaussian noise and Speckle noise on the same scene dataset. (a) Matching results of Gaussian noise with different variances. (b) Matching results of Speckle noise with different variances.

in multimodal RSIs. The notable antinoise performance can be attributed to the following factors. First, the proposed Unet-like backbone network excels in extracting multiscale features with robustness, thereby providing feature representations with high distinguishability. Second, the combination of CNN-based local features and the transformer attention mechanism could effectively improve the adaptability to multimodal RSI matching.

## VI. Conclusion

This article has presented a robust multimodal RSI matching method called MICM. Our method combines CNN and transformer attention mechanism to obtain highly reliable learning features and consists of three key modules. First, a Unet-like feature extraction backbone network is employed to learn multiscale invariant features from RSIs with different modalities. This backbone network extracts rich and evenly distributed features. Second, a hybrid approach is proposed, which combines CNN local features with the transformer attention mechanism. It effectively aggregates local learning features, enabling the capture of fine details while facilitating long-range modeling. This combination enhances the representation ability of the features. Third, a feature consistency correlation strategy is employed to maximize the number of correct corresponding keypoints, which ensures reliable matching performance by establishing consistent correspondences. Overall, the proposed MICM method offers a stable solution for multimodal RSI matching by leveraging advanced techniques in feature extraction, representation, and correspondence. The qualitative and quantitative experiments conducted on both the same scene and different scene datasets demonstrate the superior performance of our MICM compared to commonly used and state-of-the-art handcrafted- and learning-based methods. The MICM method consistently achieves better results in terms of the number of correct matches, accuracy, and robustness. These results highlight the potential of the MICM method as a promising approach for addressing multimodal RSI matching problems. The proposed method provides a valuable reference for improving the accuracy and efficiency of multimodal RSI matching tasks.

However, the method faces the challenge of relatively high computational complexity, resulting in increased runtime. To better realize its application potential, we will focus on improving the matching efficiency. We will further improve matching efficiency to balance the robustness, accuracy, and efficiency in the multimodal RSI matching tasks.

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